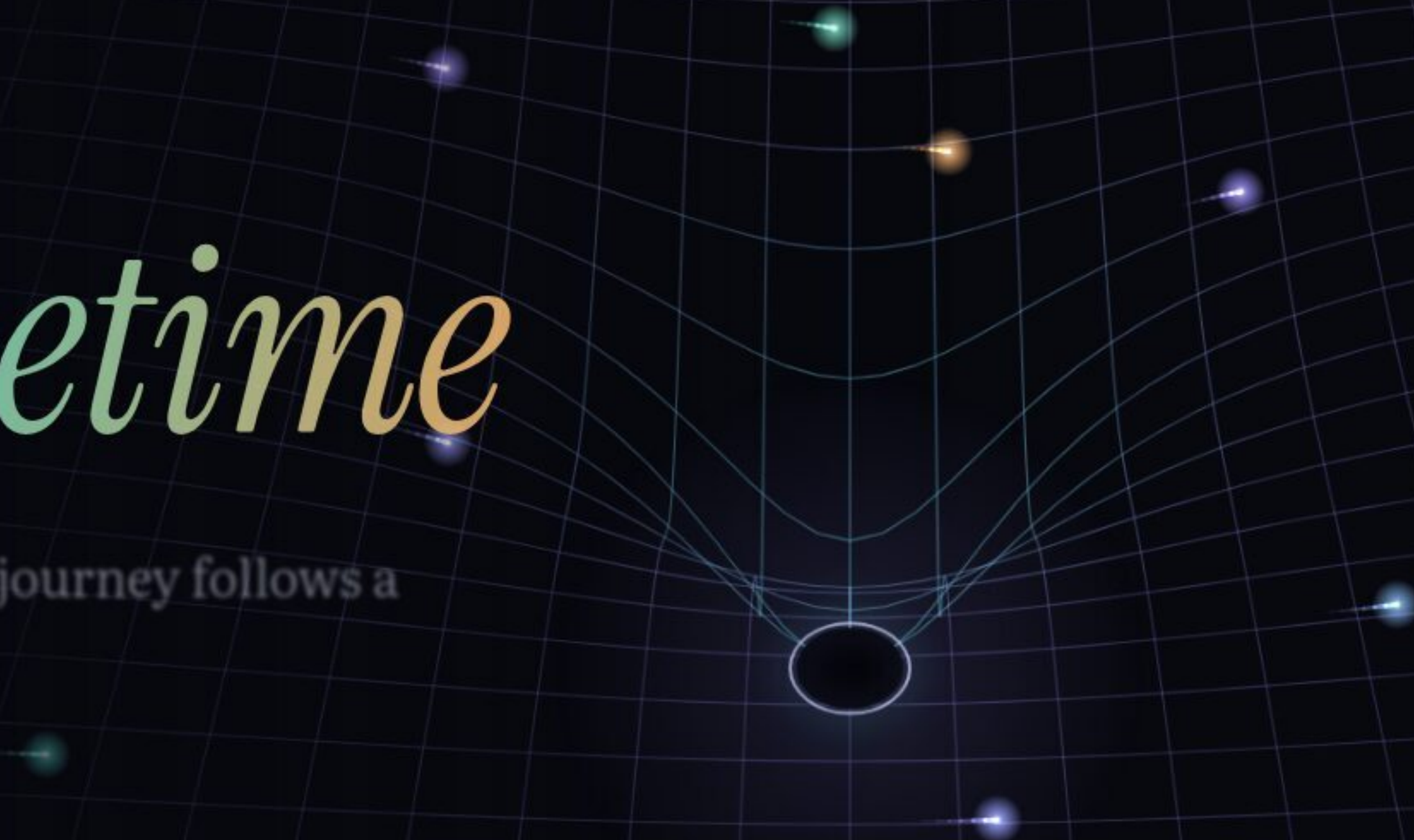


— SEMINAR

Neutrinos in *curved spacetime*

What happens to flavour oscillation when the journey follows a geodesic — and not a straight line.



Three parts, *one question*

I

Curved spacetime

Einstein, Schwarzschild, compact objects: where only General Relativity describes things well, and what the metric does to a neutrino.

II

The supernova, and neutrinos in the well

How the star collapses, where the shock wave comes from, and what happens to flavour when the emission takes place deep in a gravitational well.

III

GHOST: results

The shock front at the MSW resonance: the criterion, the cost in P_H , the bias at DUNE and where to look.

MSW: a *two-level* problem

$$H = \frac{1}{2} \begin{pmatrix} V - \Delta \cos 2\theta & \Delta \sin 2\theta \\ \Delta \sin 2\theta & -(V - \Delta \cos 2\theta) \end{pmatrix}$$

EFFECTIVE HAMILTONIAN, TWO FLAVOURS

$$\Delta = \frac{\Delta m^2}{2E}$$

$$V(r) = \sqrt{2} G_F n_e(r)$$

The resonance happens when the diagonal term vanishes:

$$V = \Delta \cos 2\theta$$

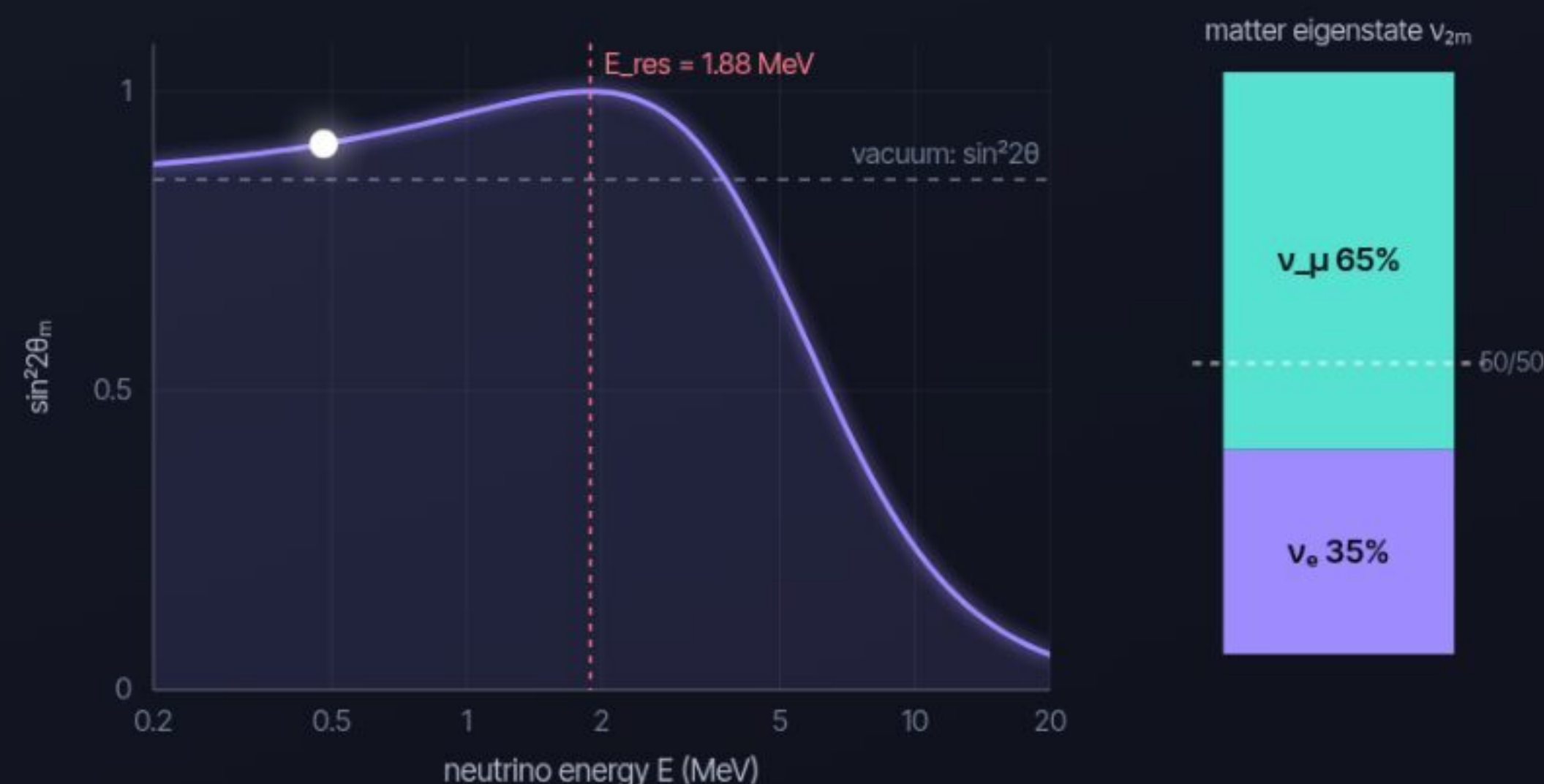
$$H_{\text{res}} \sim \frac{1}{2} \begin{pmatrix} 0 & \Delta \sin 2\theta \\ \Delta \sin 2\theta & 0 \end{pmatrix}$$

OFF-DIAGONAL ONLY: 50/50 EIGENSTATES — MAXIMAL MIXING

An **avoided crossing** with adiabatic evolution — and the bridge to supernovae, where the profile $n_e(r)$ sweeps through the resonance.

$$H = \frac{1}{2} \begin{pmatrix} -22.35 & +72.10 \\ +72.10 & +22.35 \end{pmatrix}$$

values in 10^{-12} eV
pink: $V - \Delta \cos 2\theta$ · turquoise: $\Delta \sin 2\theta$
medium: solar core, $\rho Y_e = 100 \text{ g/cm}^3$
 $V = 7.63 \cdot \Delta = 78.08$



E 0.48 MeV $V - \Delta \cos 2\theta$ -22.35 $\Delta \sin 2\theta$ +72.10 θ_m 36.4° $\sin^2 2\theta_m$ 0.912

In the basis that *follows the matter*

Change from the flavour basis to the instantaneous matter basis:

$$\begin{pmatrix} \nu_e \\ \nu_x \end{pmatrix} = U(\theta_m(x)) \begin{pmatrix} \nu_1^m \\ \nu_2^m \end{pmatrix}$$

Since U depends on x , the Schrödinger equation picks up a term:

$$i \frac{d}{dx} \begin{pmatrix} \nu_1^m \\ \nu_2^m \end{pmatrix} = \left[\begin{pmatrix} -\Delta_m/2 & 0 \\ 0 & +\Delta_m/2 \end{pmatrix} - i U^\dagger \frac{dU}{dx} \right] \begin{pmatrix} \nu_1^m \\ \nu_2^m \end{pmatrix}$$

$$-i U^\dagger \frac{dU}{dx} \sim \begin{pmatrix} 0 & i d\theta_m/dx \\ -i d\theta_m/dx & 0 \end{pmatrix}$$

Bridge to the previous slide: $U(\theta_m)$ diagonalizes the H there:

$$\Delta_m \cos 2\theta_m = \Delta \cos 2\theta - V \cdot \Delta_m \sin 2\theta_m = \Delta \sin 2\theta$$

The previous slide gives the Hamiltonian at one point; diagonalizing it defines θ_m and Δ_m ; here we ask what happens when θ_m changes along the path.

The density gradient reappears as an **off-diagonal** term in the basis of instantaneous eigenstates.

A head-to-head competition:

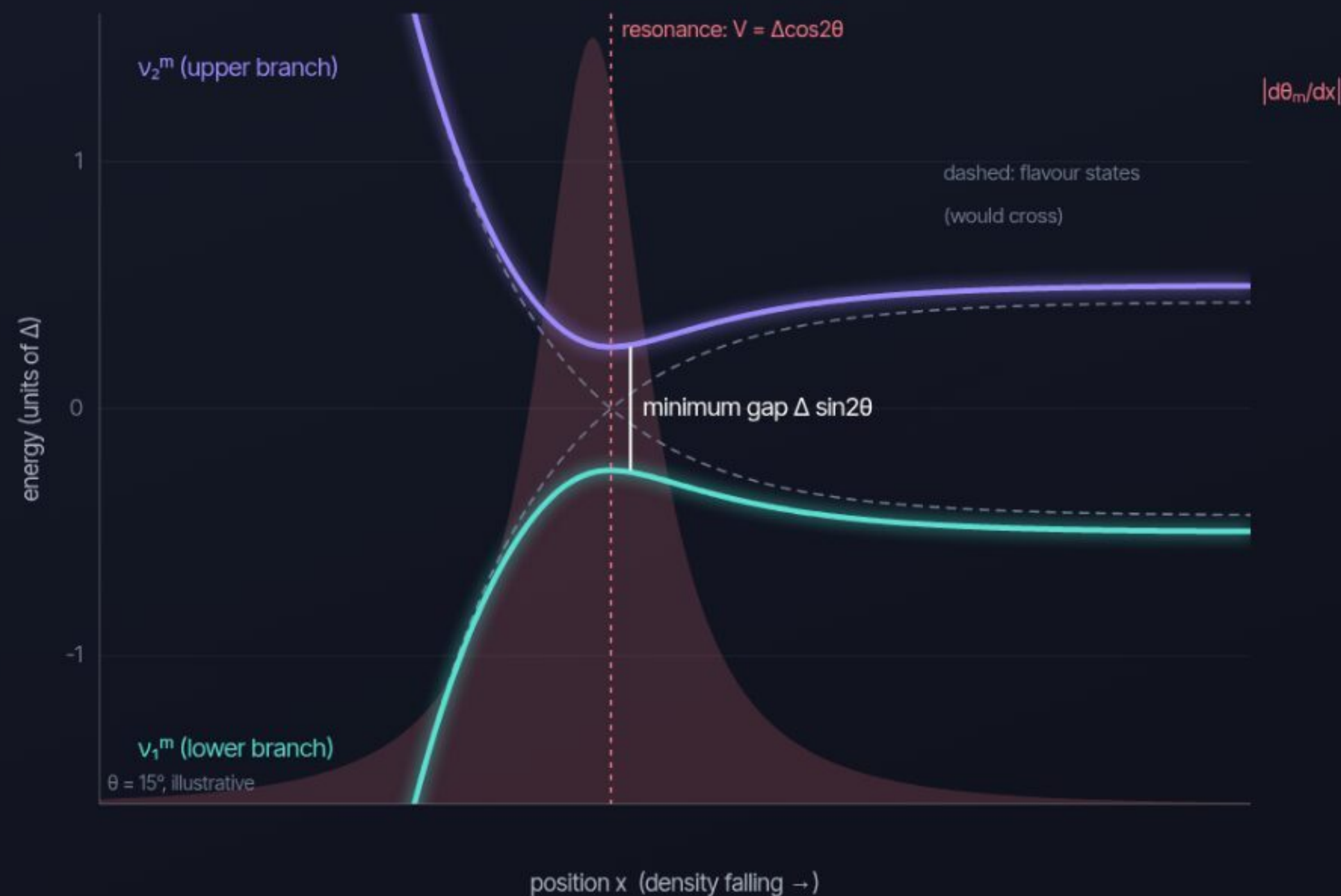
$$\Delta_m \text{ versus } |d\theta_m/dx|$$

Adiabatic condition:

$$|d\theta_m/dx| \ll \Delta_m$$

More fundamental than “the density varies slowly”: what matters is how fast the basis rotates compared with the gap.

Why the resonance is *the critical place*



At resonance, $V = \Delta \cos 2\theta$ and $\theta_m = \pi/4$. The level separation is smallest:

$$\Delta_m^{\text{res}} = \Delta \sin 2\theta$$

An **avoided crossing**. Far from it the levels are well apart; near it they come as close as they can — and that is where the basis turns fastest (pink area).

That is where the $\nu_1^m \leftrightarrow \nu_2^m$ transition is easiest. So adiabaticity is evaluated **at the resonance**.

The adiabaticity parameter γ and *Landau–Zener*

$$\gamma_{\text{res}} = \frac{\Delta m^2}{2E} \frac{\sin^2 2\theta}{\cos 2\theta} \left| \frac{d \ln n_e}{dx} \right|^{-1}_{\text{res}}$$

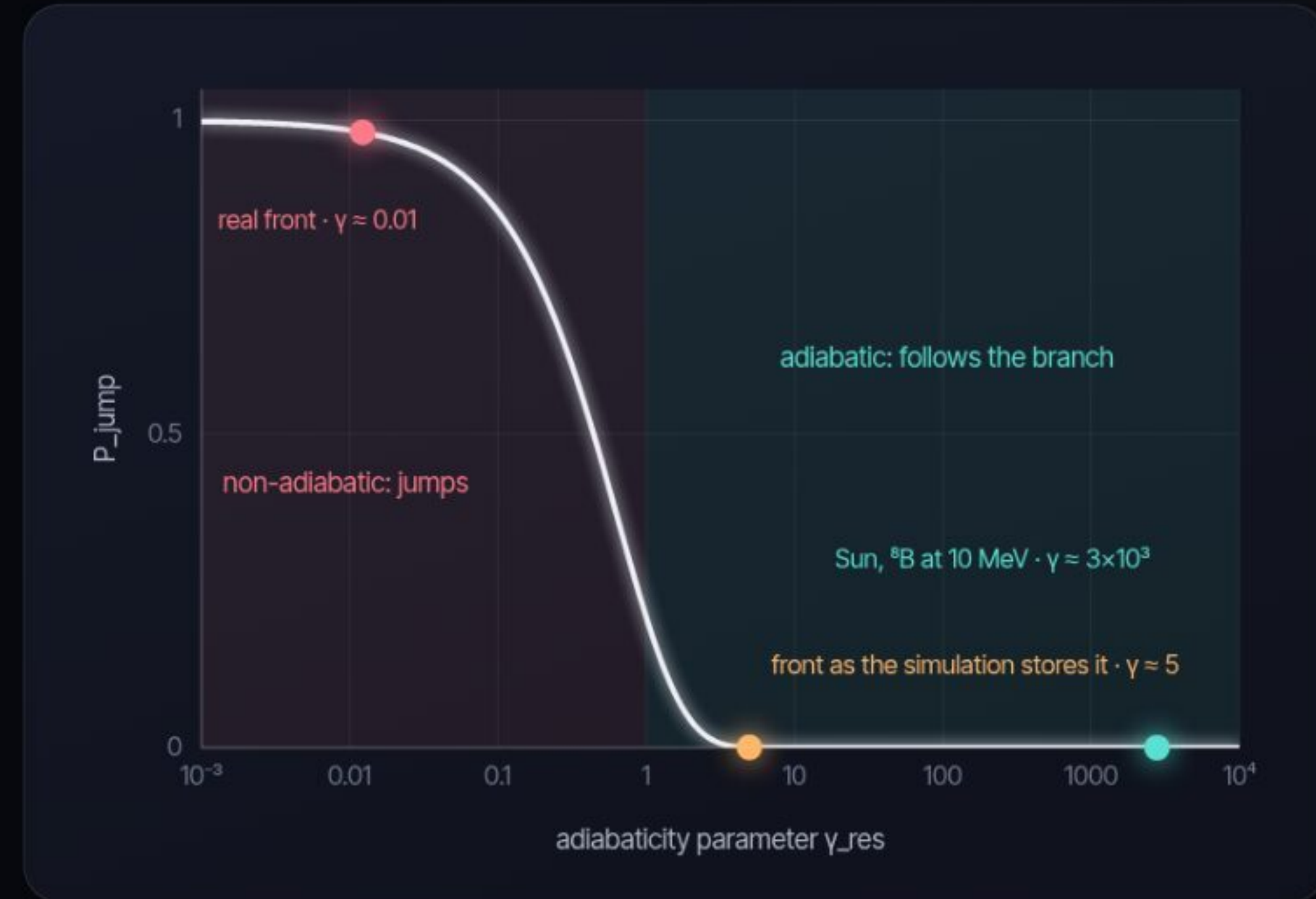
ADIABATICITY PARAMETER = GAP ÷ ROTATION OF THE BASIS, AT RESONANCE

$$P_{\text{jump}} \simeq e^{-\pi \gamma_{\text{res}}/2}$$

PROBABILITY OF JUMPING BRANCH (LANDAU–ZENER, NEARLY LINEAR PROFILE)

IF...	γ	P_{jump}	WHY
larger energy E	↓	↑	$\Delta = \Delta m^2/2E$ drops: smaller gap
larger Δm^2	↑	↓	larger gap
larger mixing θ (up to 45°)	↑	↓	larger minimum gap $\Delta \sin 2\theta$
density varies slowly	↑	↓	scale $ d \ln n_e/dx ^{-1}$ large: the basis turns slowly
shock front (step)	→ 0	max.	scale → 0: sudden jump

$\gamma \gg 1 \rightarrow$ adiabatic: follows the branch · $\gamma \lesssim 1 \rightarrow$ jumps branch



The resonance determines where the eigenstates mix maximally; adiabaticity determines whether the neutrino follows those eigenstates as they change.

The shock front *crosses the resonance*



A large, stylized, light blue letter 'I' that serves as a logo for the series.

PART ONE

Curved *spacetime*

Gravity is not a force: it is geometry. Far from stars, the difference is a tiny correction. Next to a compact object, it is everything.

Einstein · 1915

Schwarzschild · 1916

compact objects

gravitational waves

collapsars

HR1403

Taurus

67 Tauri

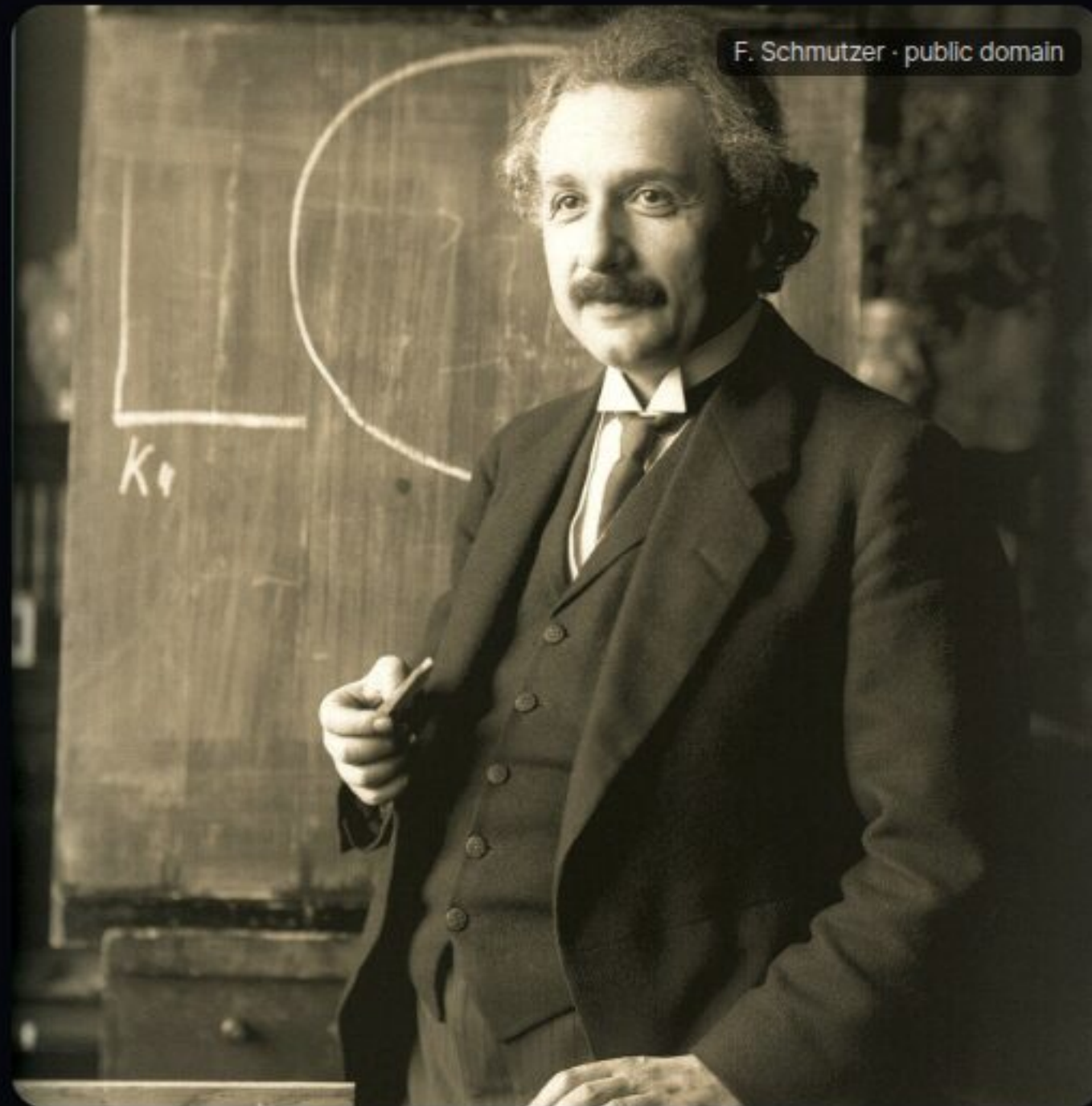
65 Tauri

72 Tauri

Eclipse of 29 May 1919 · ESO / Landessternwarte Heidelberg · CC BY 4.0

— 25 NOVEMBER 1915

Einstein and *his field equations*



Albert Einstein

1921

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

CURVATURE = ENERGY AND MOMENTUM CONTENT

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = 0$$

GEODESIC

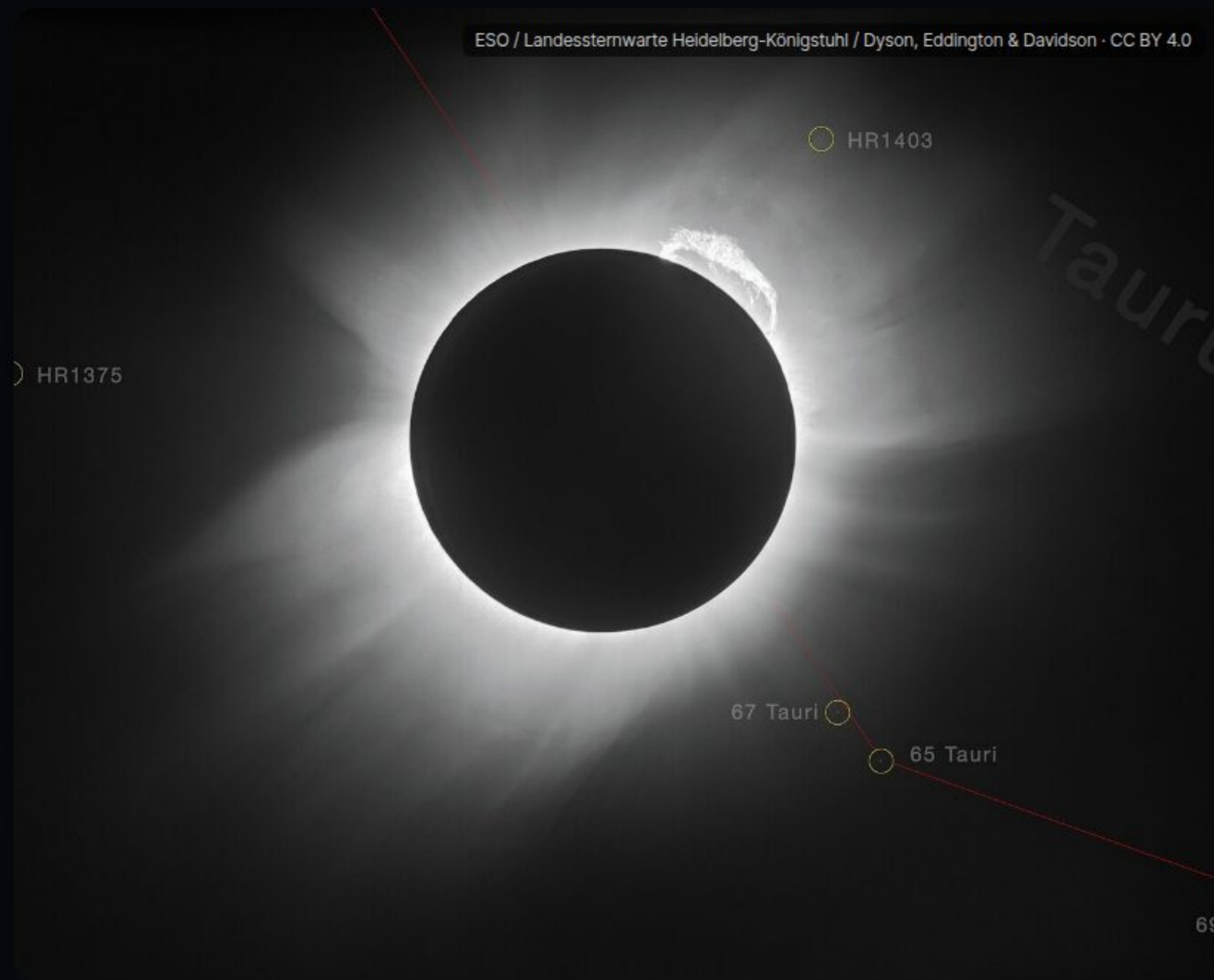
$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

THE METRIC IS THE FIELD

“Spacetime tells matter how to move; matter tells spacetime how to curve.”

John A. Wheeler

Sobral, 1919: light *bends*



EFFECT	GR PREDICTS
perihelion of Mercury (1915)	43" / century
light grazing the Sun (1919)	1.75"
redshift, Pound–Rebka (1959)	2.5×10^{-15} over 22.5 m
Shapiro delay (1964)	~200 μ s
GPS clocks	+38 μ s / day

Sobral expedition (Ceará, Brazil): **$1.98 \pm 0.12''$** . Príncipe: $1.61 \pm 0.30''$.
The sky over north-eastern Brazil confirmed Einstein.

Mercury's orbit *does not close*



PERIHELION ADVANCE	PER CENTURY
observed (relative to the stars)	574"
other planets, by Newton	531"
unexplained remainder	43"

$$\Delta\varphi = \frac{6\pi GM_{\odot}}{c^2 a(1-e^2)} = 0,104'' \text{ per orbit}$$

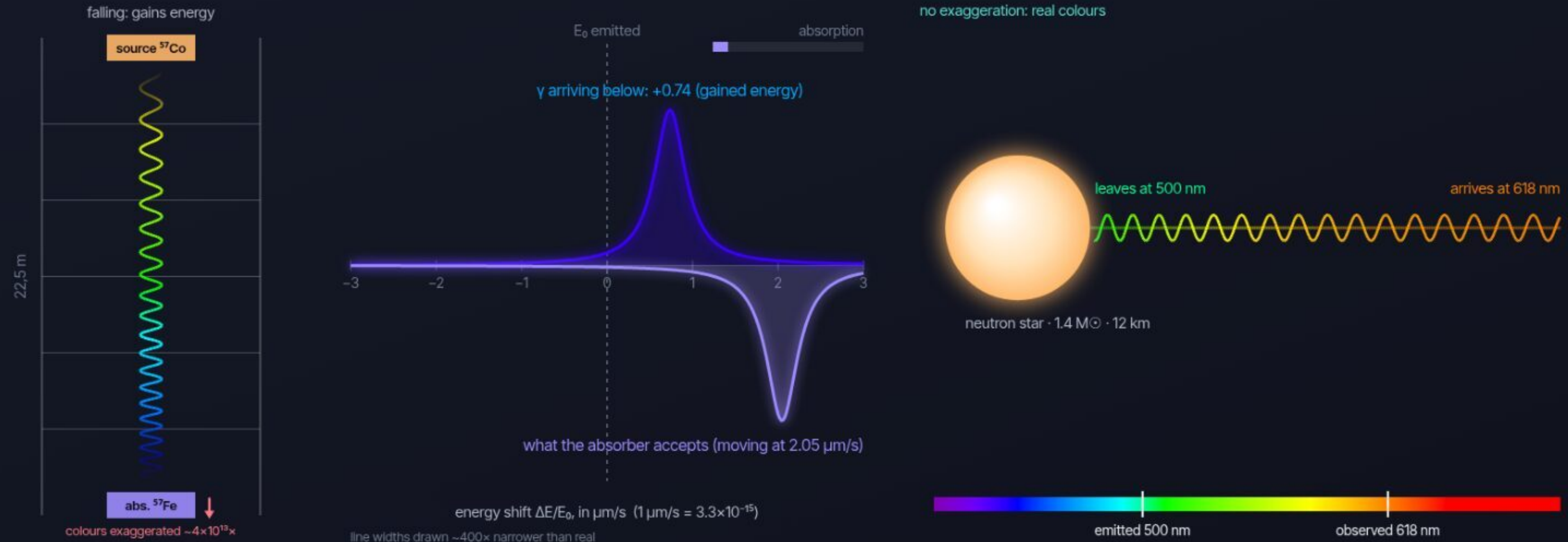
GR · EINSTEIN, NOVEMBER 1915 · 415 ORBITS PER CENTURY → 43.0"

Le Verrier found the remainder in 1859 and proposed a hidden planet, Vulcan. It was never found. Einstein got the 43" with no adjustable parameter.

"For a few days, I was beside myself with joyous excitement."

Einstein to Ehrenfest, January 1916

Light *loses energy as it climbs*



tower $\Delta v/v = gh/c^2$ 2.46×10^{-15}

resonance at $+0.74 \mu\text{m/s} \approx 2.7 \text{ mm per hour}$

absorber now $2.05 \mu\text{m/s}$

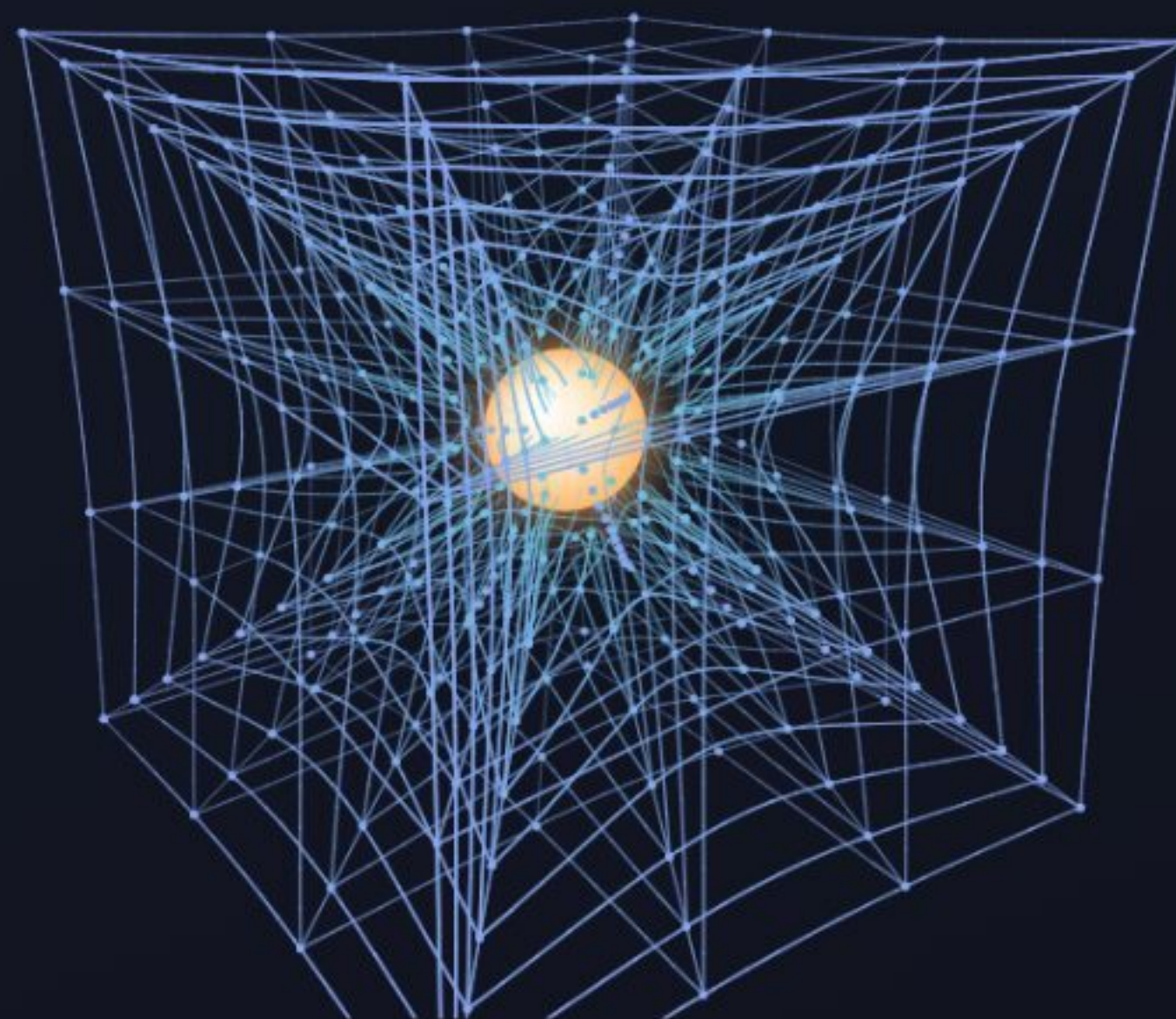
star $1+z = (1-r_g/R)^{-1/2}$ 1.235

14.4 keV γ rays from ^{57}Fe (Mössbauer effect). Pound & Rebka, PRL 4, 337 (1960): $\sim 10\%$; Pound & Snider, Phys. Rev. 140, B788 (1965): $\sim 1\%$. Every 12 s source and absorber swap places, as in the experiment.

Space falls · *neutron star*

• SCHWARZSCHILD IN GULLSTRAND-PAINLEVÉ · $1.4 M_{\odot}$ · $R = 12 \text{ KM}$

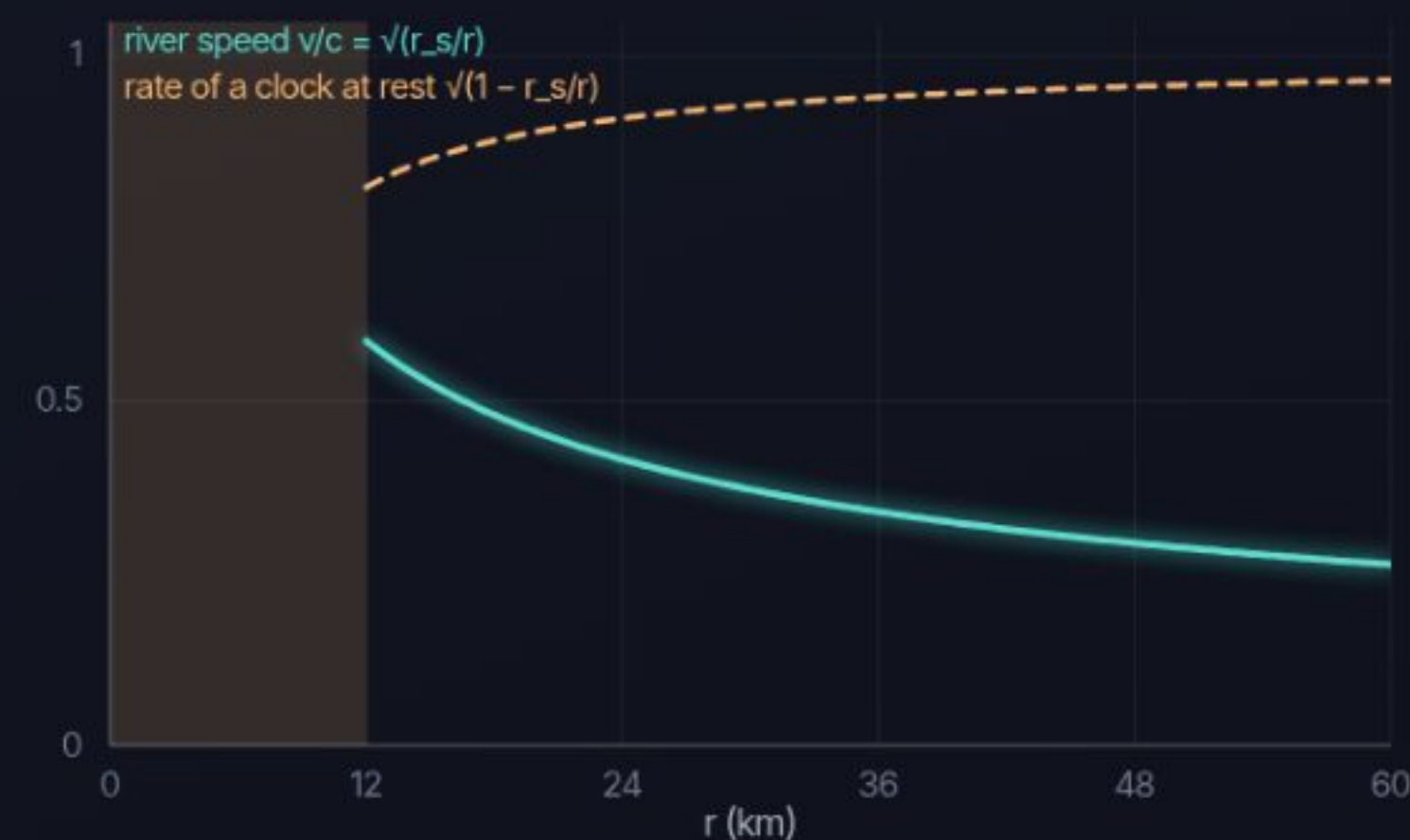
each node: a freely falling frame



r_s 4.1 km at the surface v 0.59 c fall from 60 km 463 μs cell: radial $\times 2.2$ · lateral $\times 0.20$

drag the lattice to rotate

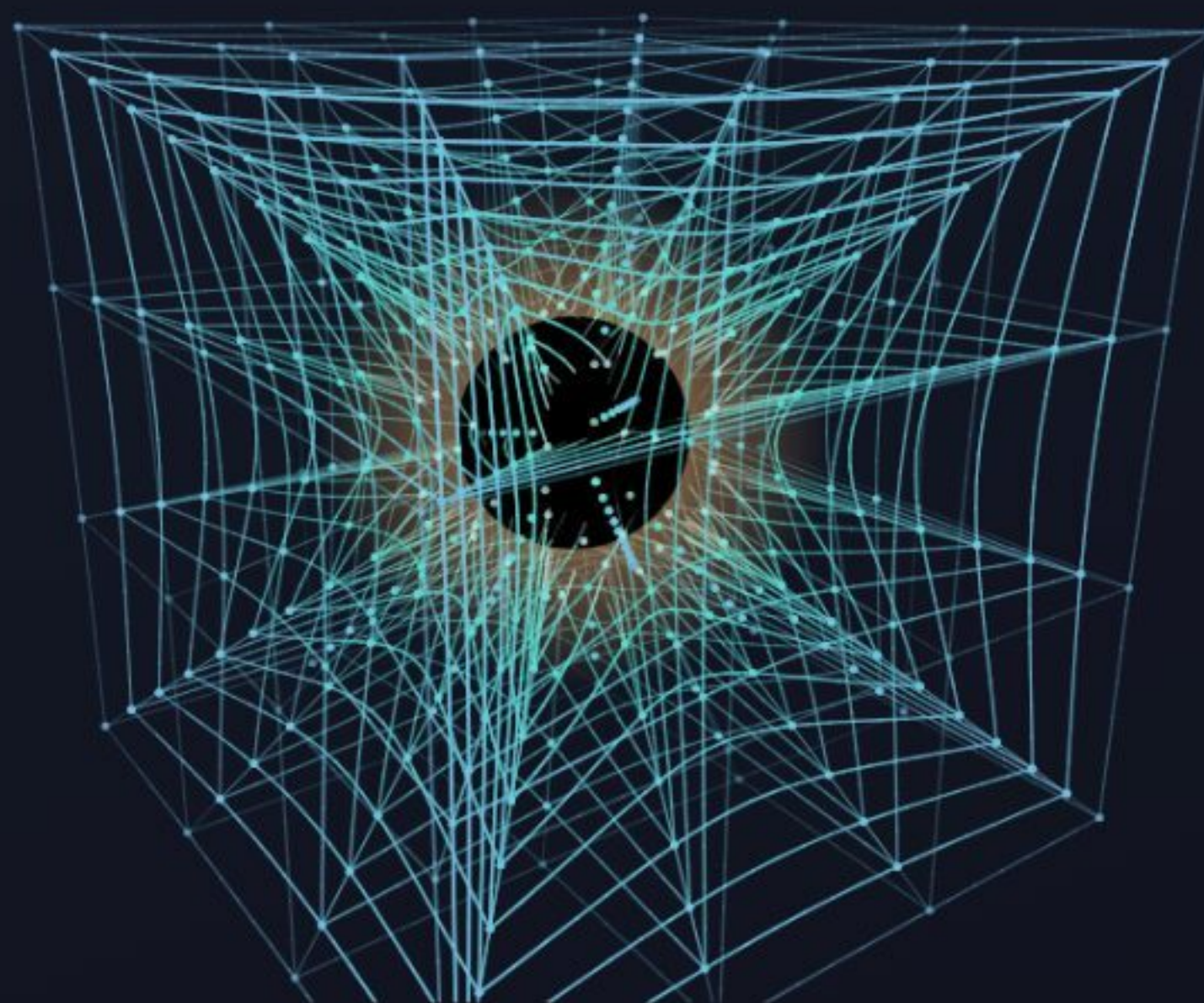
Each node is a **freely falling local frame**, released from rest far away. In Gullstrand-Painlevé coordinates space is **flat** and flows inward at the escape velocity, $v = \sqrt{2GM/r}$: the lattice is carried by the river. Near the mass the nodes run faster — the lines bend, cells stretch radially and squeeze sideways: the tide. At the star's surface the river already runs at **0.59 c**.



Space falls · *black hole*

• SCHWARZSCHILD IN GULLSTRAND-PAINLEVÉ · $6 M_{\odot}$ · HORIZON AT 17.7 KM

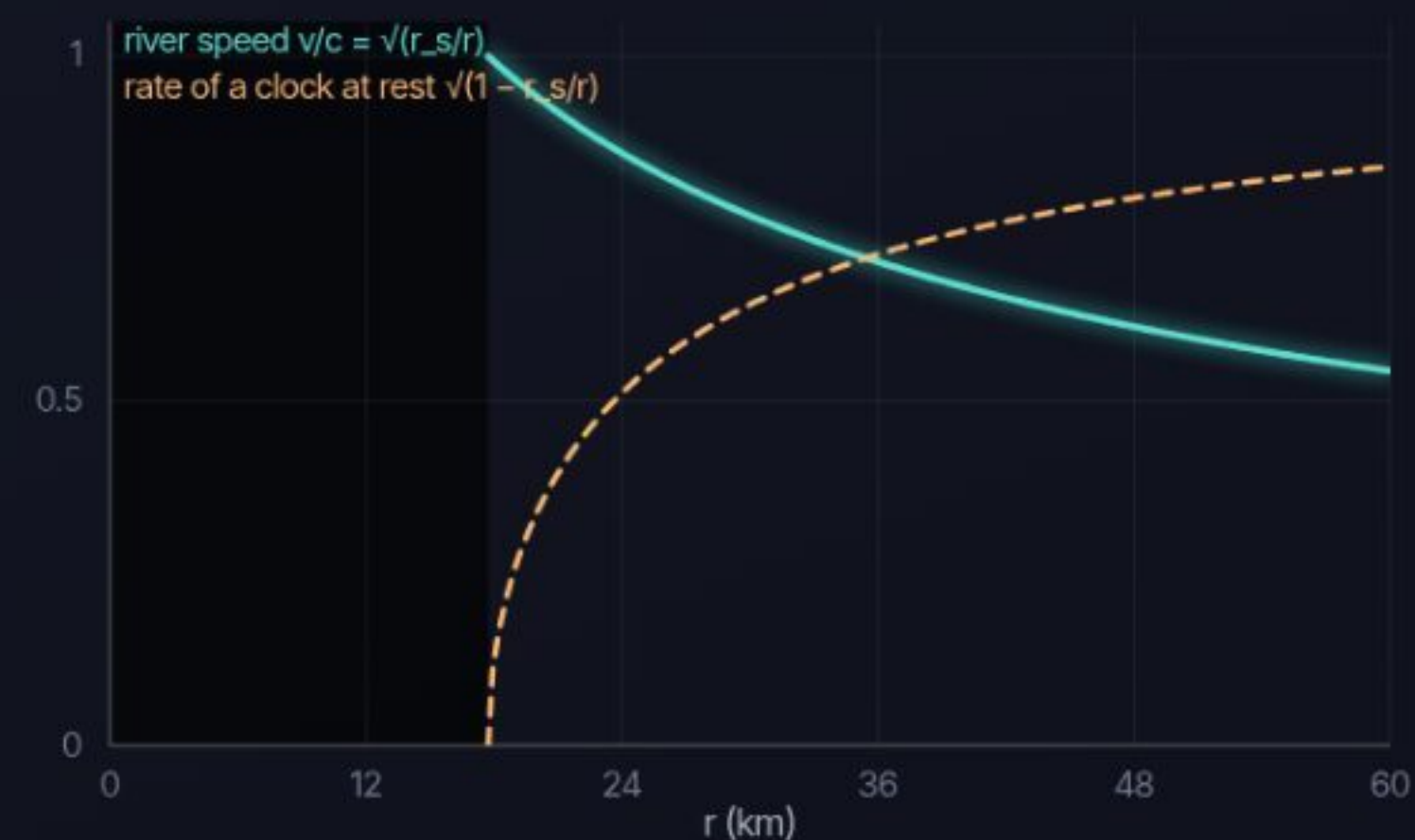
each node: a freely falling frame



r_s 17.7 km at the horizon v 1.00 c fall from 60 km 206 μ s cell: radial $\times 1.8$ · lateral $\times 0.30$

drag the lattice to rotate

The same lattice, now with no surface to stop the fall. At the **horizon** the river runs at exactly **c**: a light signal aimed outward can only stand still there, and inside it is dragged inward. That is why nothing gets out. The dashed line is the rate of a clock held at rest — it drops to **zero** at the horizon, where staying at rest is no longer possible.



Schwarzschild and the *first exact solution*



Karl Schwarzschild

1873 – 1916

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \frac{dr^2}{1 - r_s/r} + r^2 d\Omega^2$$

$R_s = 2GM/c^2$ · VALID OUTSIDE ANY NON-ROTATING SPHERICAL BODY

1 r_s

event horizon

1.5 r_s

photon sphere

3 r_s

ISCO: last stable
orbit

2.95 km

r_s of the Sun
(Earth: 8.9 mm)

He wrote to Einstein on 22 December 1915 while serving as an artillery officer. He died in May 1916.

Curvature does *two things* to a neutrino

1 · IT CHANGES THE ENERGY

A neutrino emitted at r with local energy E arrives at infinity with

$$E_{\infty} = E \sqrt{1 - r_s/r}$$

GRAVITATIONAL REDSHIFT, $R_s = 2GM/C^2$

The whole spectrum slides down, and the source's clock runs slower as seen from outside.

2 · IT BENDS THE PATH

The trajectory is a **geodesic**, not a straight line. Next to a compact object the deflection is large, and different paths can connect the same source to the same detector.

Photons and neutrinos follow the same geometry, which SN 1987A confirmed to ~0.5%.

NEUTRINOS IN CURVED SPACETIME · II EBN

Both matter because, in a supernova, emission happens **deep in the well** and detection happens **far from it**.

AND THERE IS A THIRD, SUBTLER ONE

The oscillation phase is accumulated along the geodesic, with the particle's proper time. In flat spacetime it reduces to the familiar $\Delta m^2 L / 4E$; in a curved metric it does not.

We come back to this in three slides.

None of this is optional when the density that sets the MSW resonance is measured in the local frame and the detector sits 10 kpc away.

What arrives *is not what left*

WHERE	R	$U = R_S/R$	Z
neutrinosphere, $1.4 M_\odot$	30 km	0.138	0.077
cooler neutrinosphere	50 km	0.083	0.044
cold neutron star	12 km	0.345	0.235
collapsar disk, $3 M_\odot$	3 r_s	0.333	0.225

During accretion, the spectrum DUNE sees left the star **~8% more energetic** than it arrives.

And so does the time interval: what the star lives in 1 s, we measure in 1.08 s. The neutrino light curve is read with the wrong clock if the metric is ignored.

WHY THIS MATTERS FOR THE RESONANCE

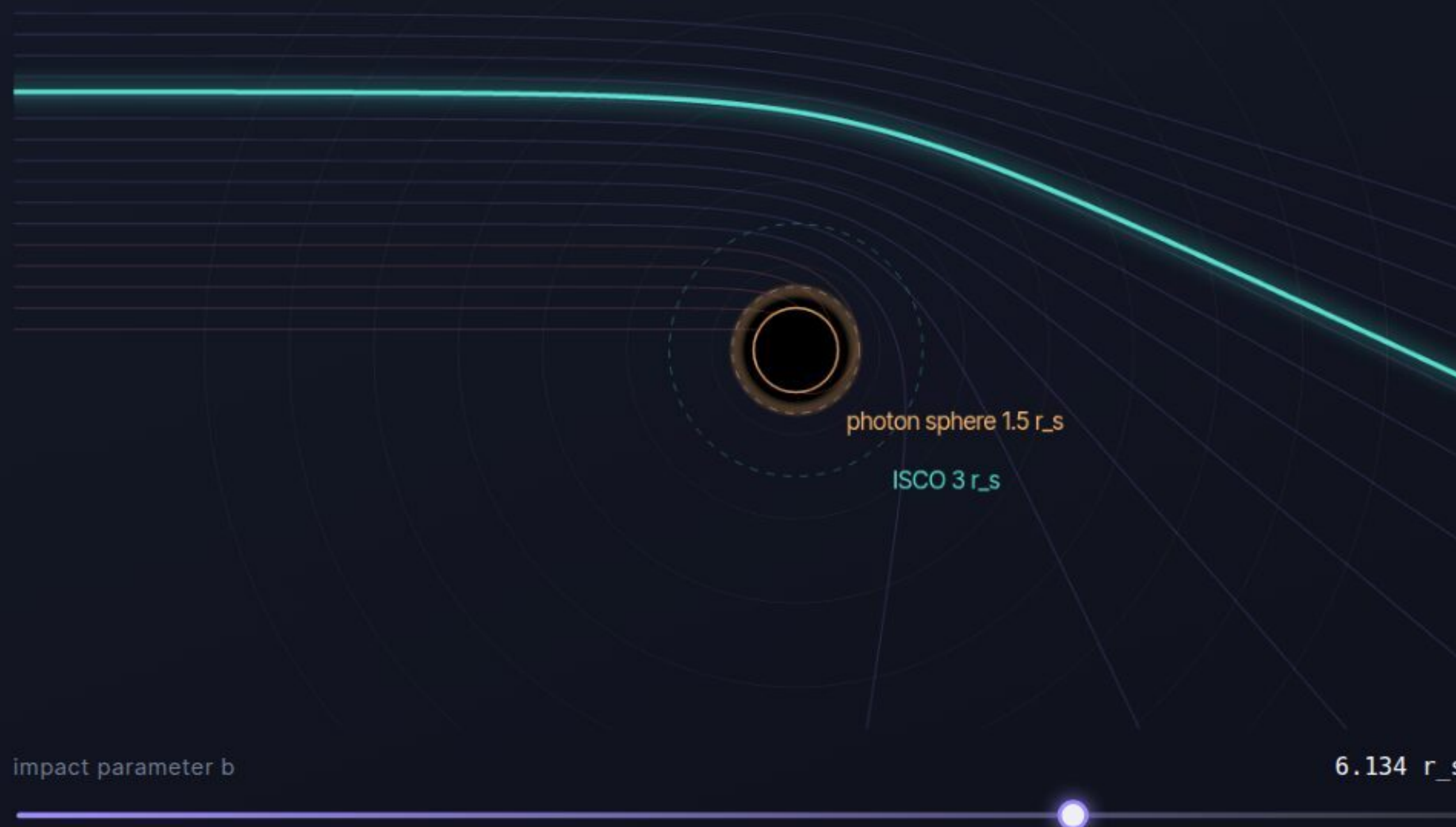
The MSW condition involves the **local** energy. Using the detected energy instead shifts the resonance density by ~8%, and therefore shifts the radius where the conversion happens.

u and z computed with $r_s = 2GM/c^2$; $z = (1 - u)^{-1/2} - 1$.

Light and neutrinos *near a black hole*

• INTERACTIVE · RK4 IN THE BROWSER

$$u'' + u = (3/2) r_s u^2, u = 1/r$$



Every ray arriving with $b < 3\sqrt{3} M$ falls in. Just above that, it loops around the photon sphere before escaping.

closest approach $5.55 r_s$

total deflection 25.4°

full loops 0

b_c $2.598 r_s$

An ultra-relativistic neutrino follows, to first order, the same null geodesic. Mass enters as a correction $\propto m^2/E^2$ — and that correction is exactly what carries the oscillation phase.

Where GR *stops* being a correction

1/8 · EARTH

Earth

Sun

Sirius B

proto-NS

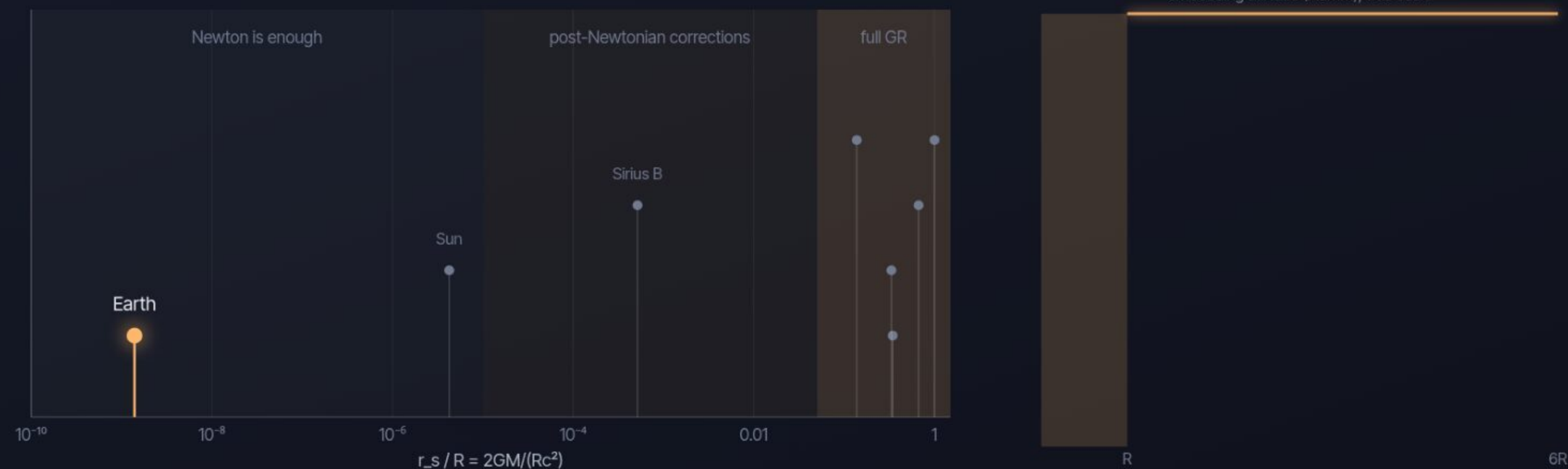
neutron star

ISCO

photon sph.

horizon

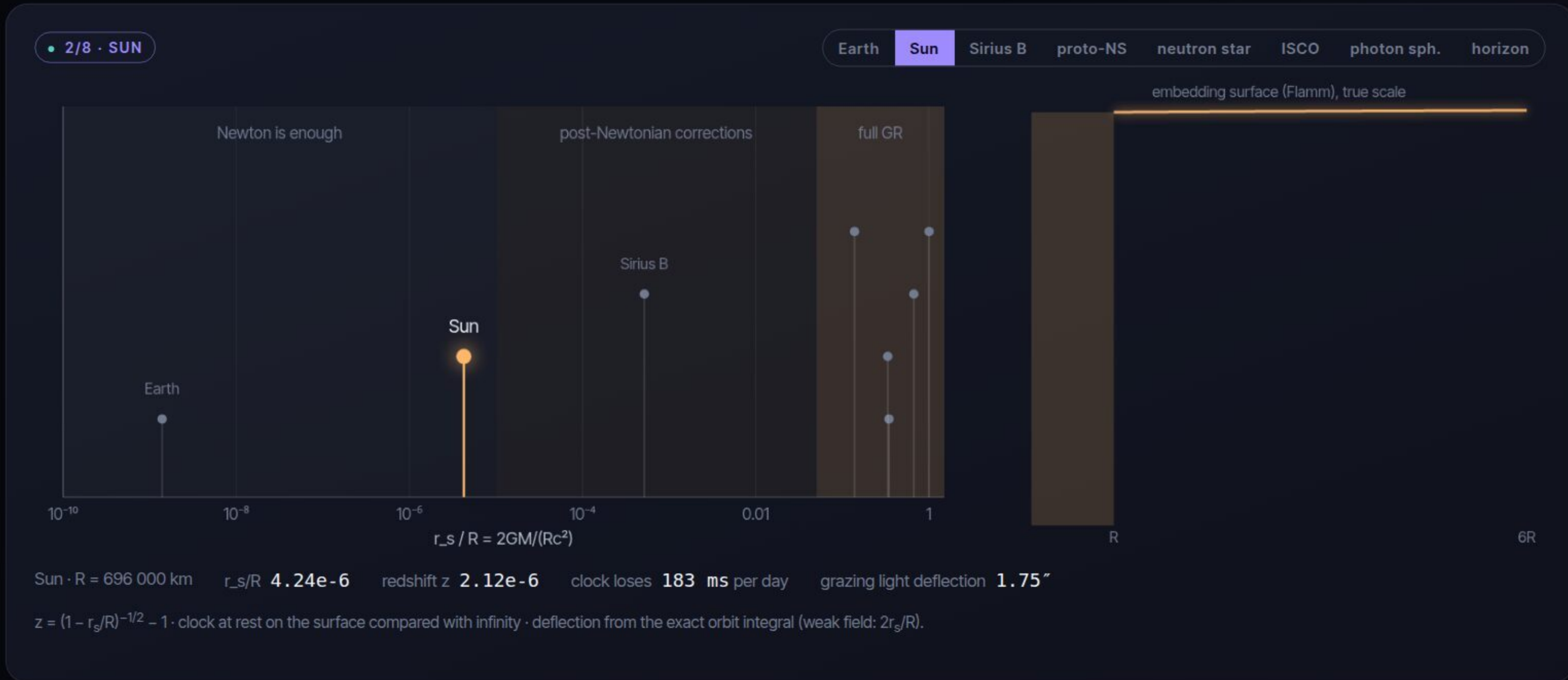
embedding surface (Flamm), true scale



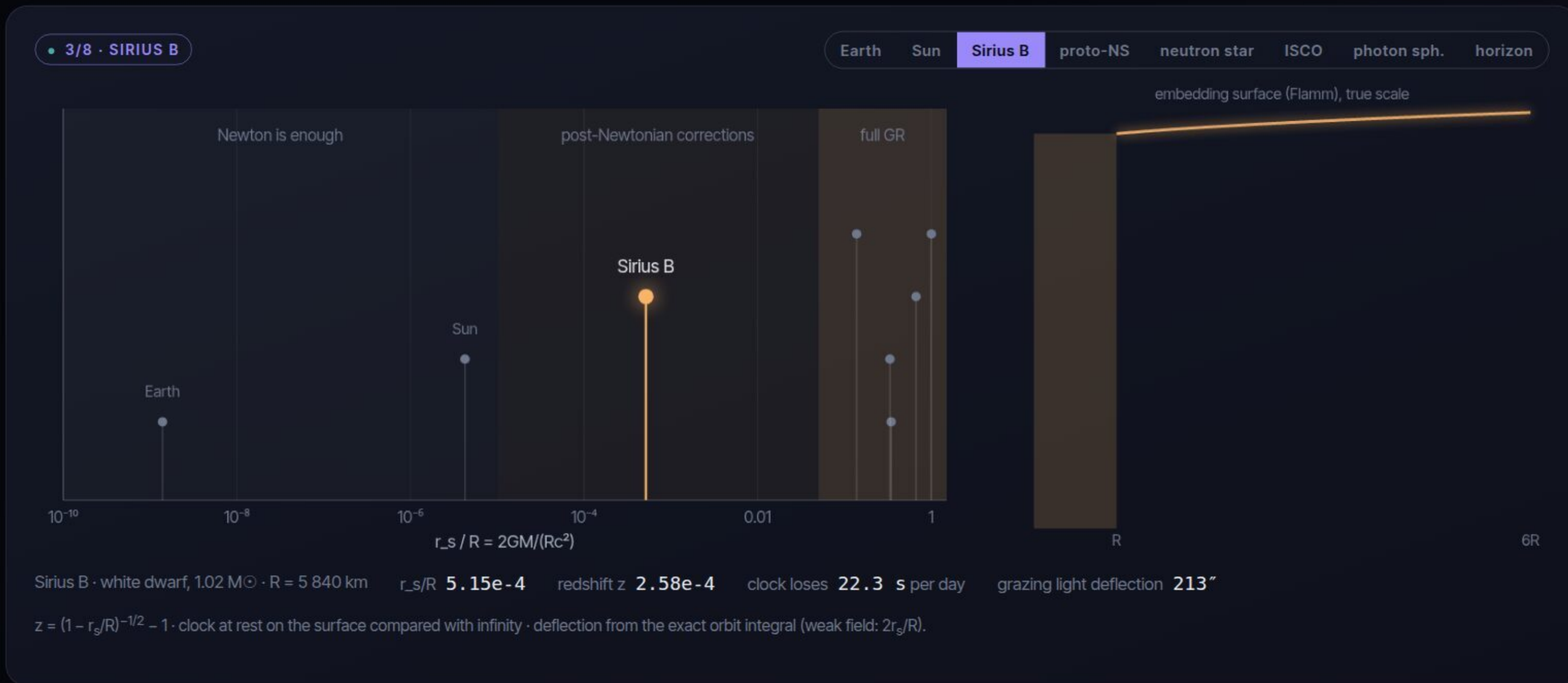
Earth · $R = 6\,371\text{ km}$ r_s/R **1.39e-9** redshift z **6.96e-10** clock loses **60.1 μs** per day grazing light deflection **0.000574"**

$z = (1 - r_s/R)^{-1/2} - 1$ · clock at rest on the surface compared with infinity · deflection from the exact orbit integral (weak field: $2r_s/R$).

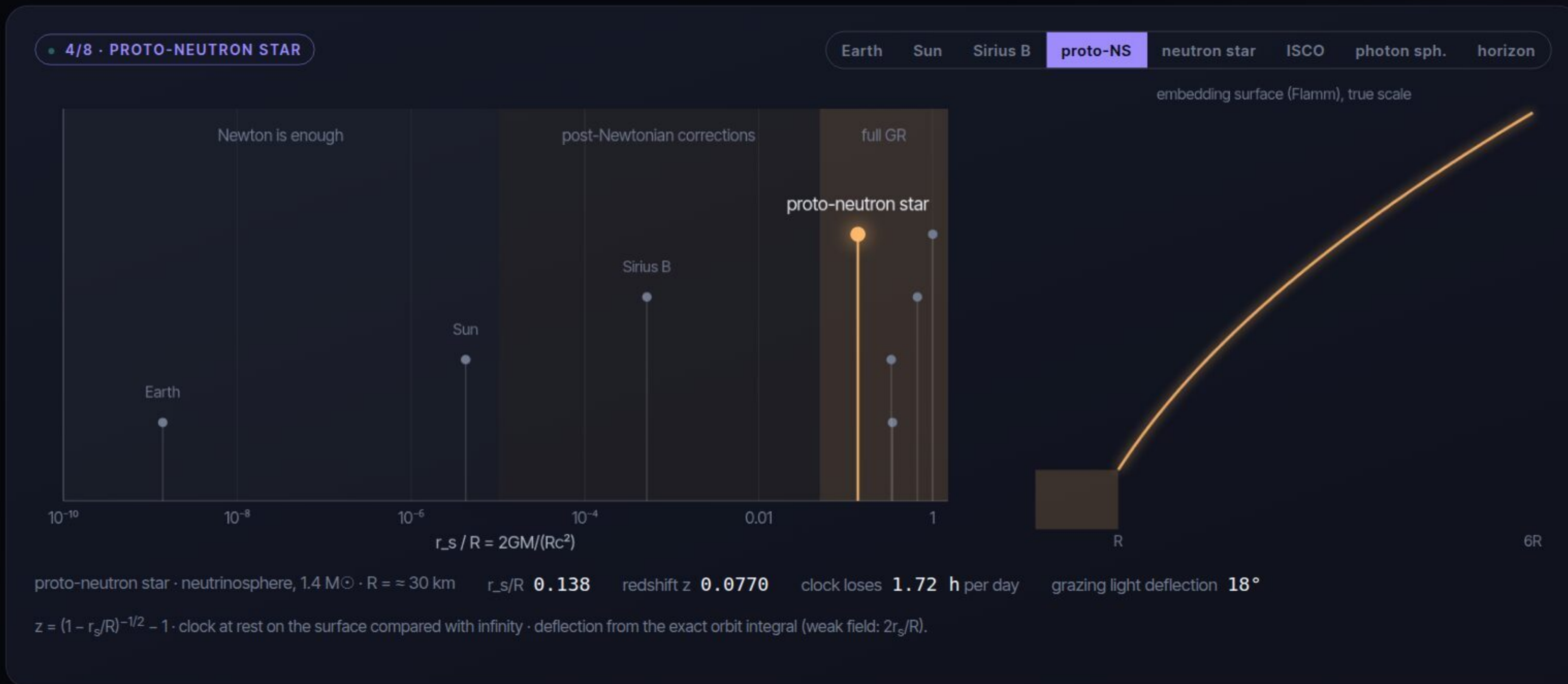
Where GR *stops* being a correction



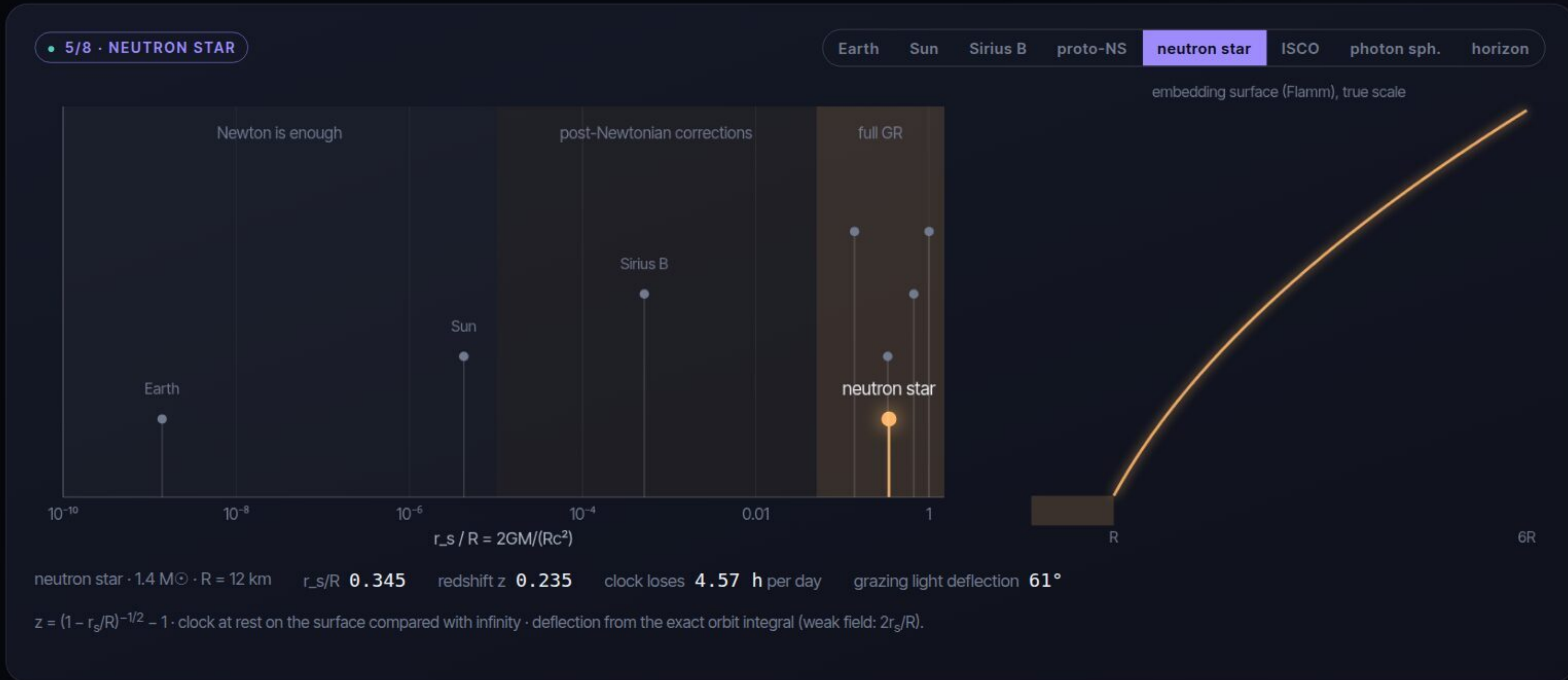
Where GR *stops* being a correction



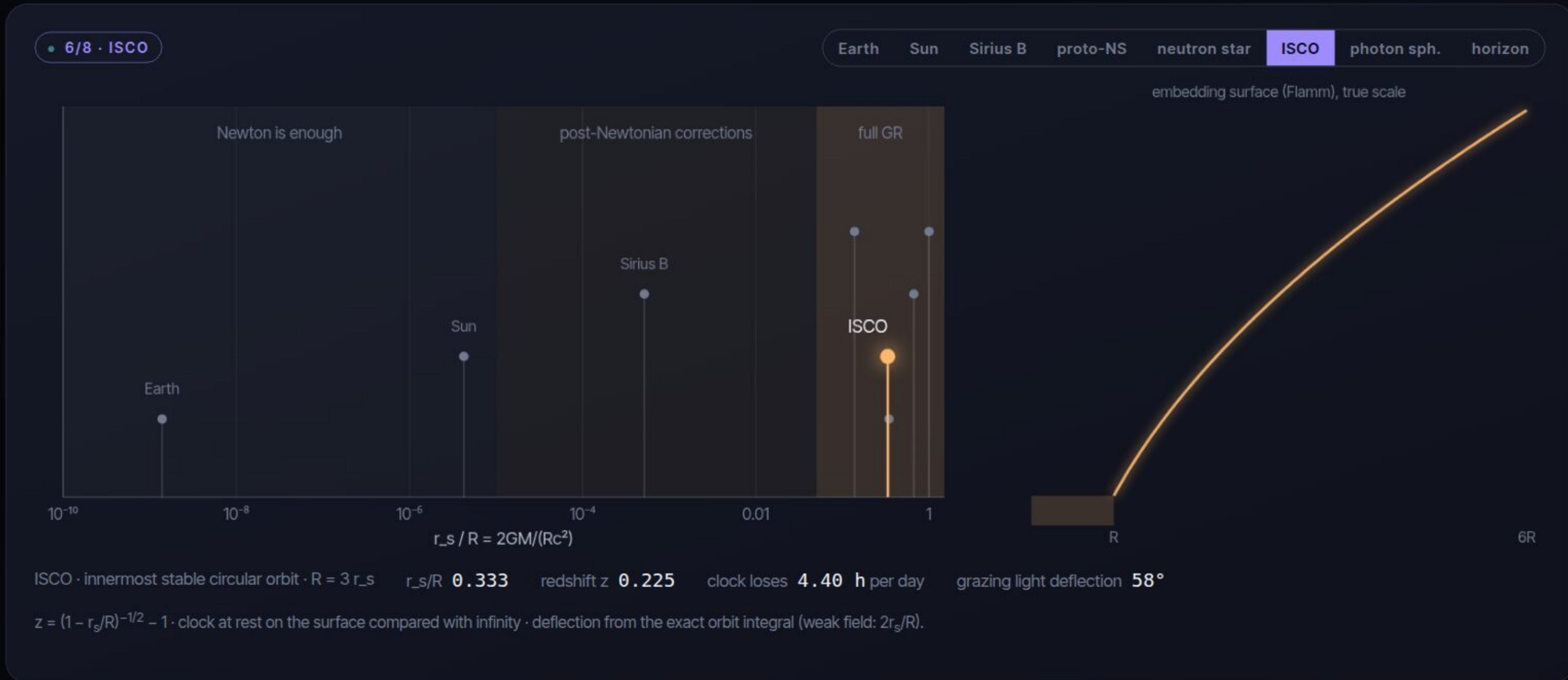
Where GR *stops* being a correction



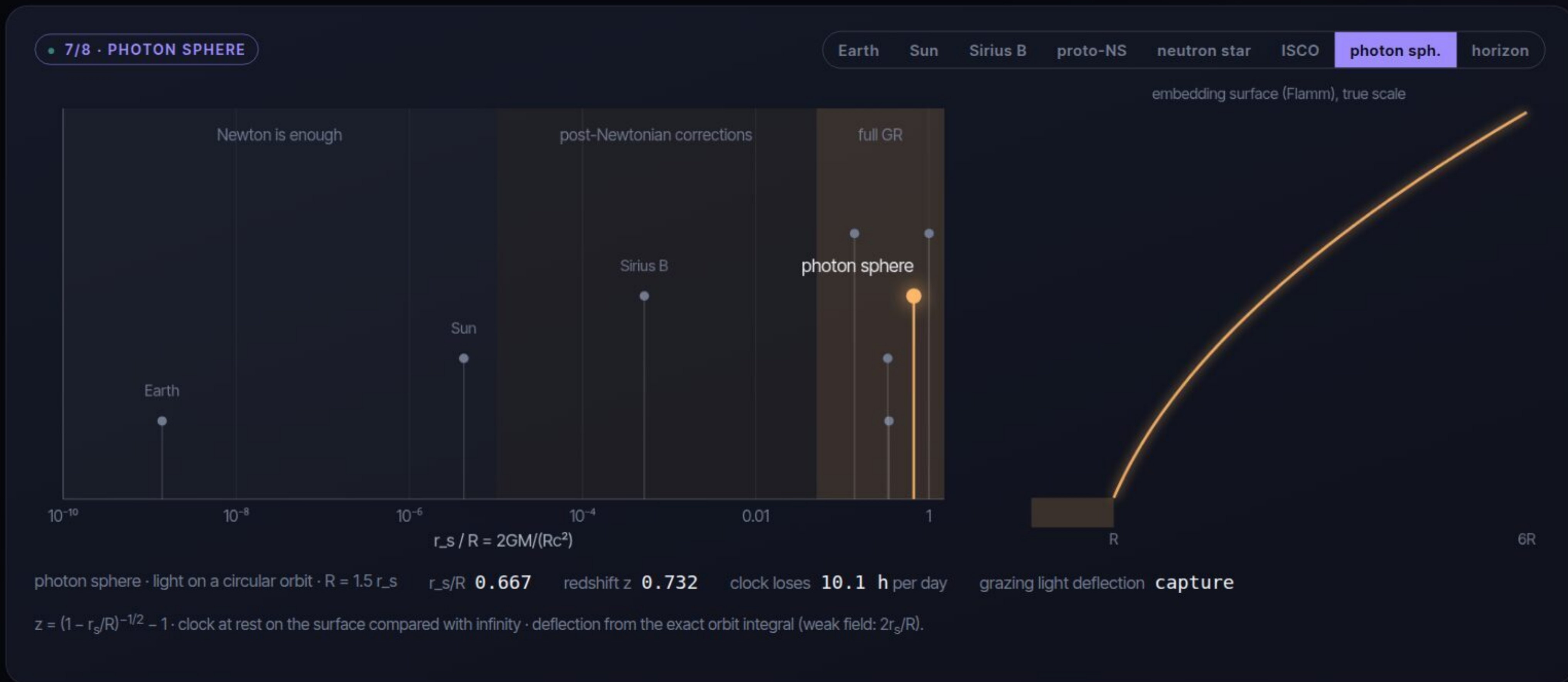
Where GR *stops* being a correction



Where GR *stops* being a correction



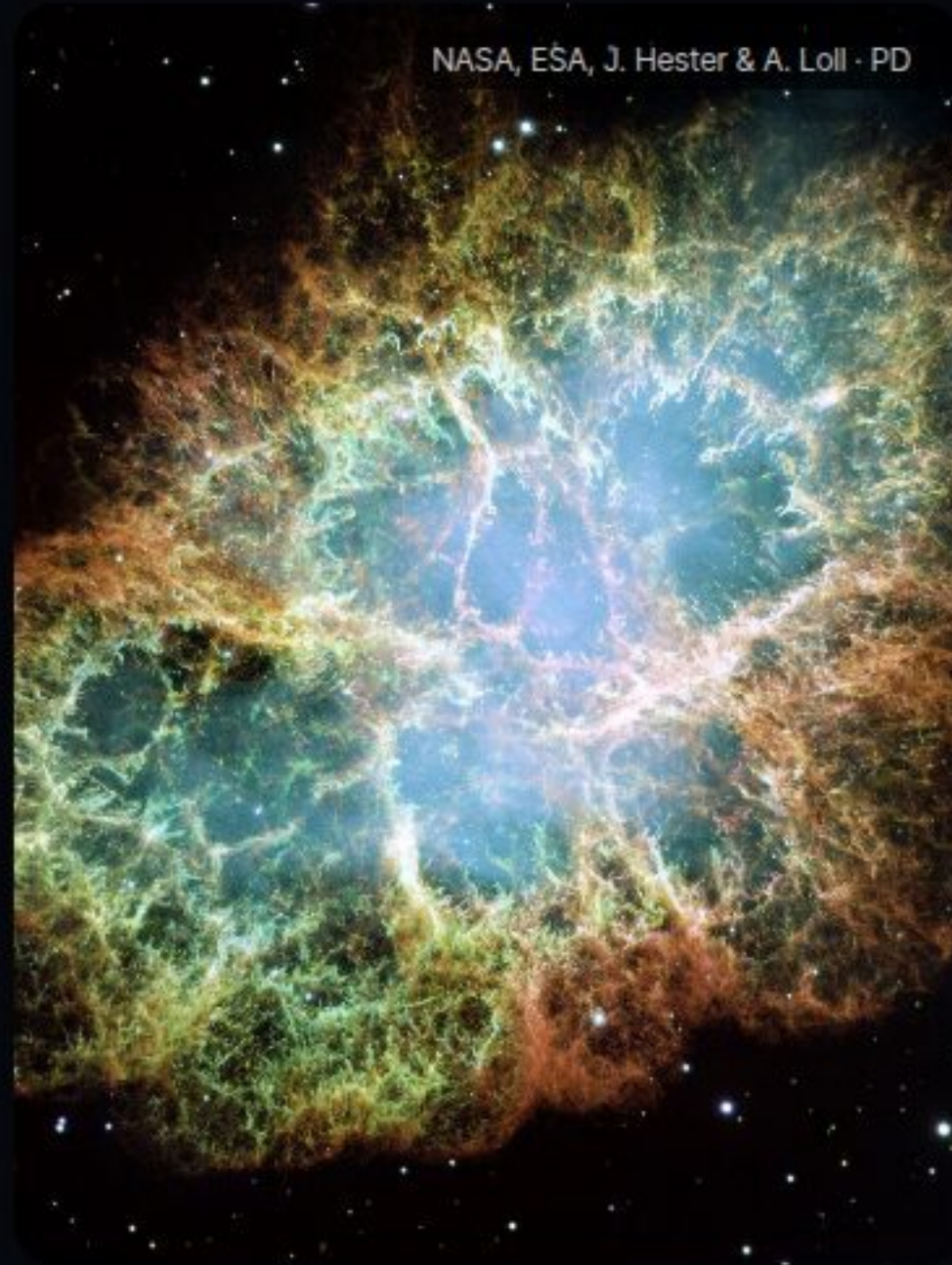
Where GR *stops* being a correction



Where GR *stops* being a correction

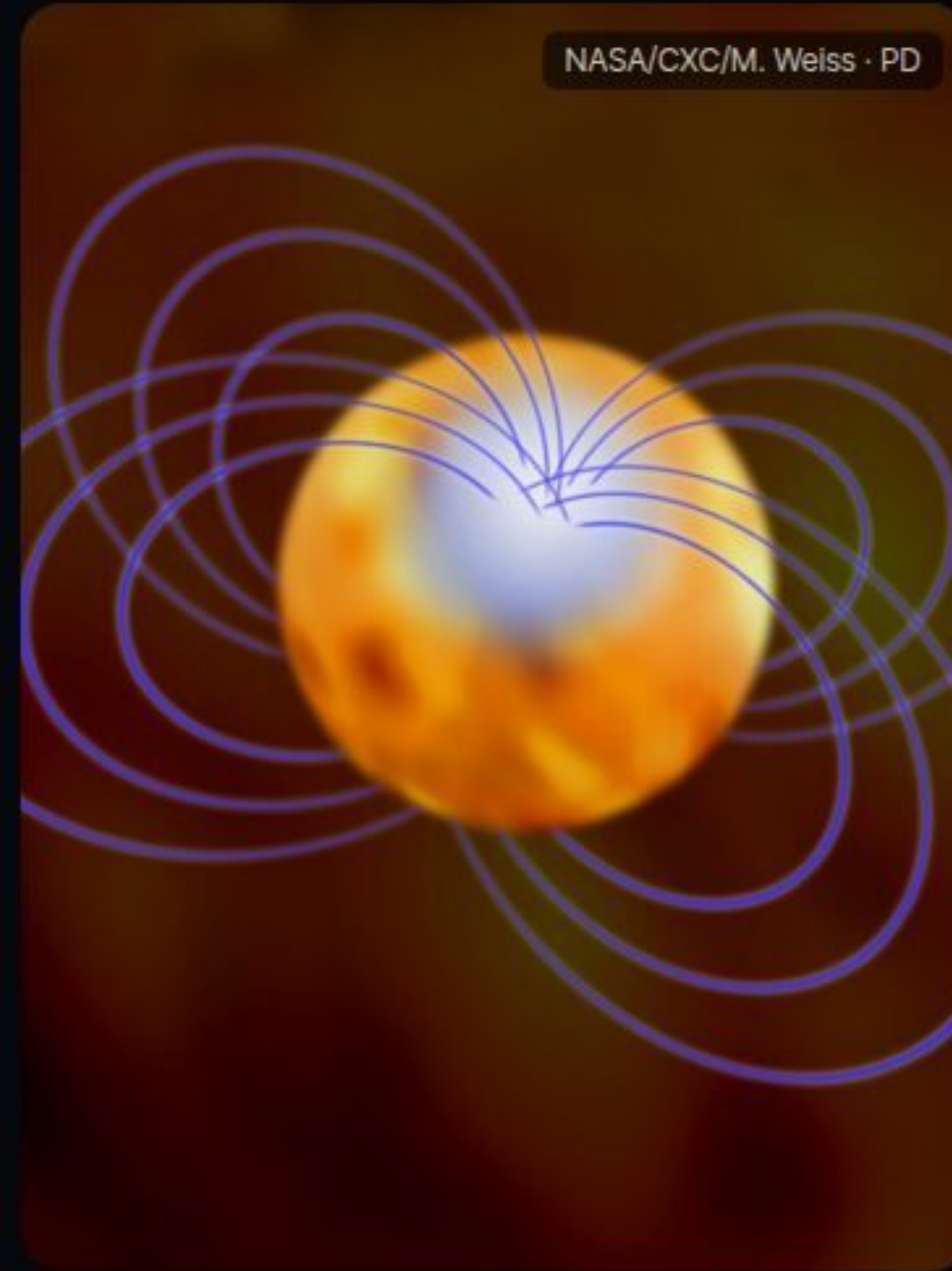


The zoo where *only GR works*



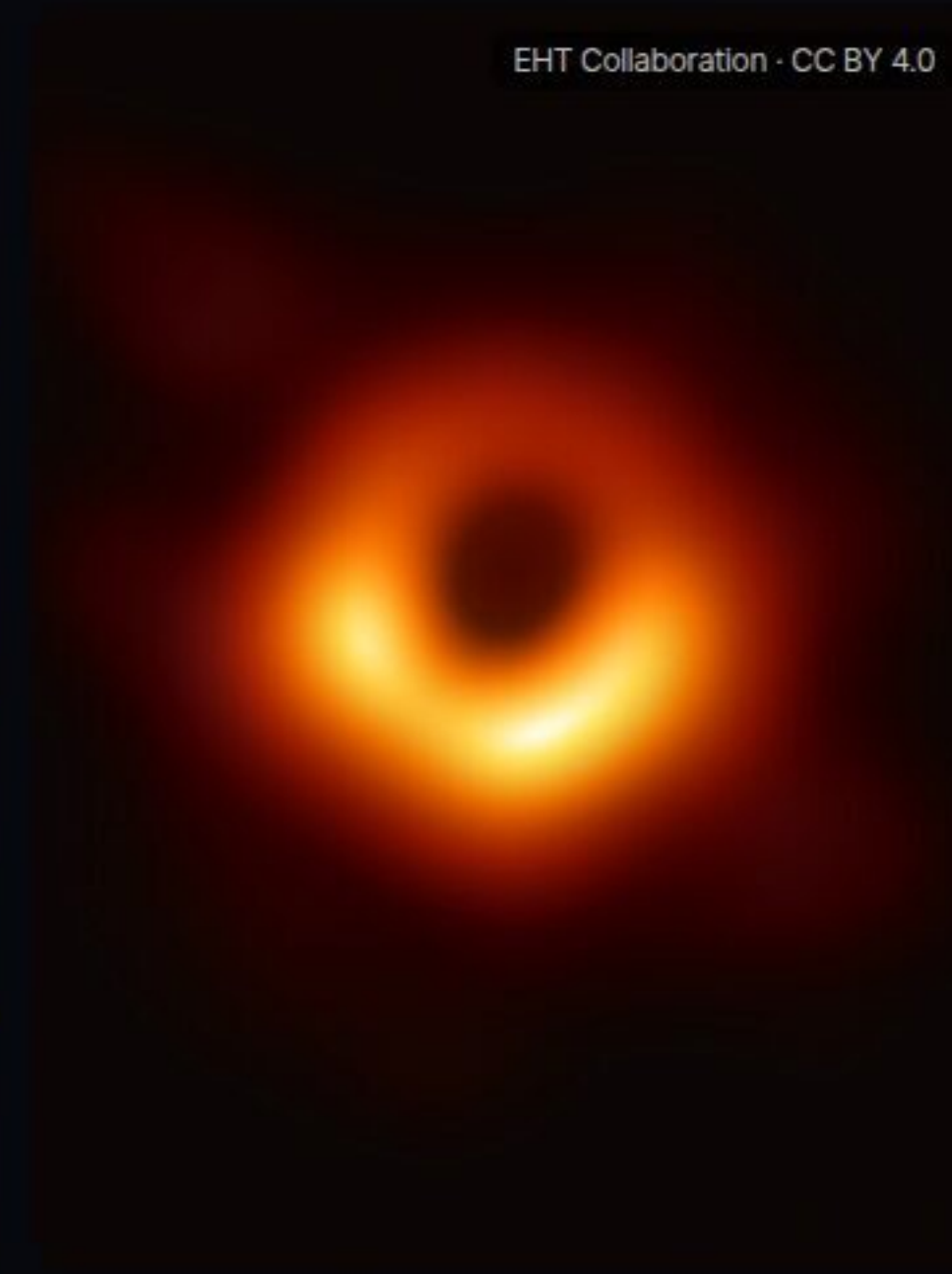
NASA, ESA, J. Hester & A. Loll · PD

Crab — remnant of the supernova of 1054, with a 33 ms pulsar at its centre.



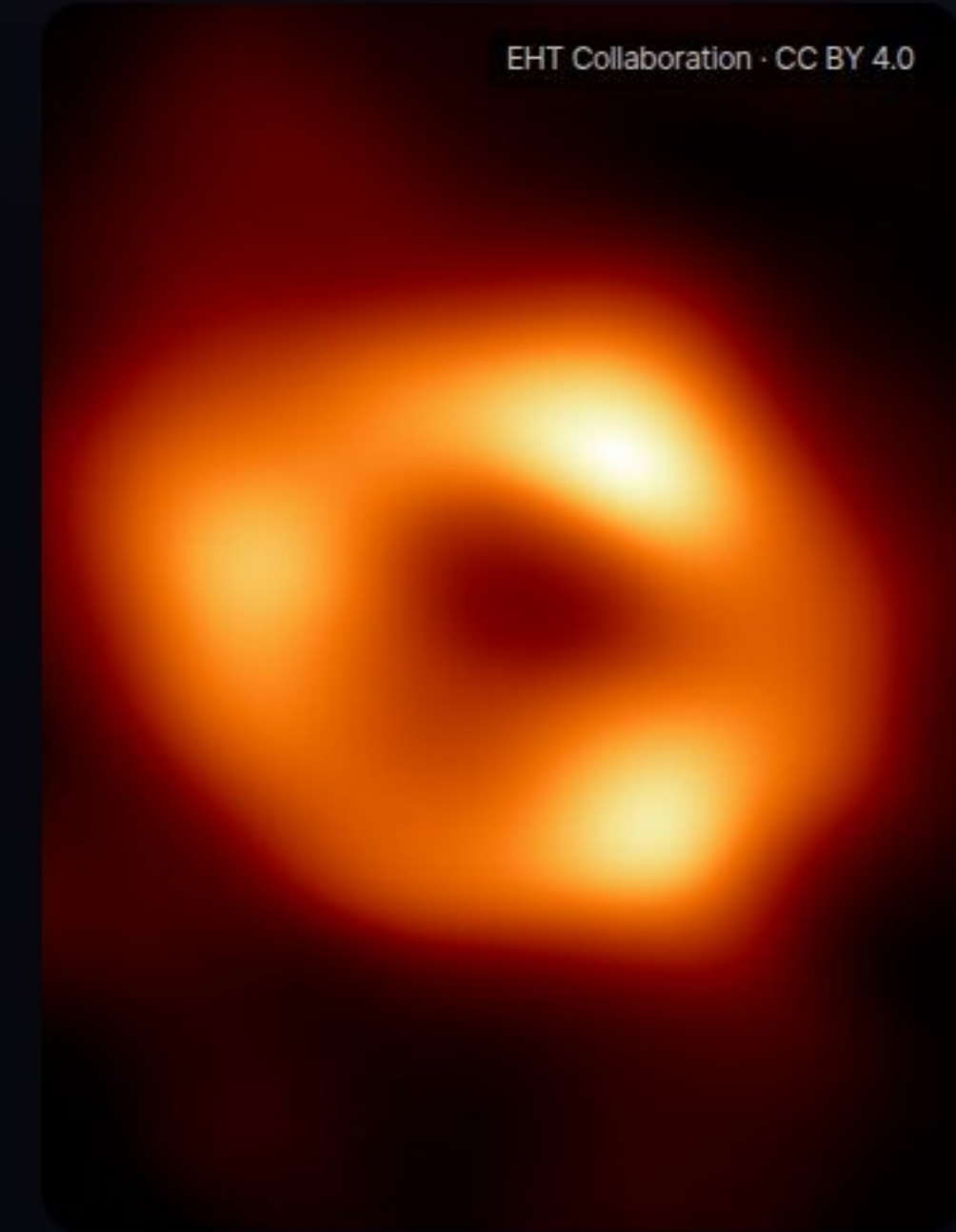
NASA/CXC/M. Weiss · PD

Neutron star — $1.4 M_{\odot}$ in 12 km, central density of several 10^{14} g/cm^3 .



EHT Collaboration · CC BY 4.0

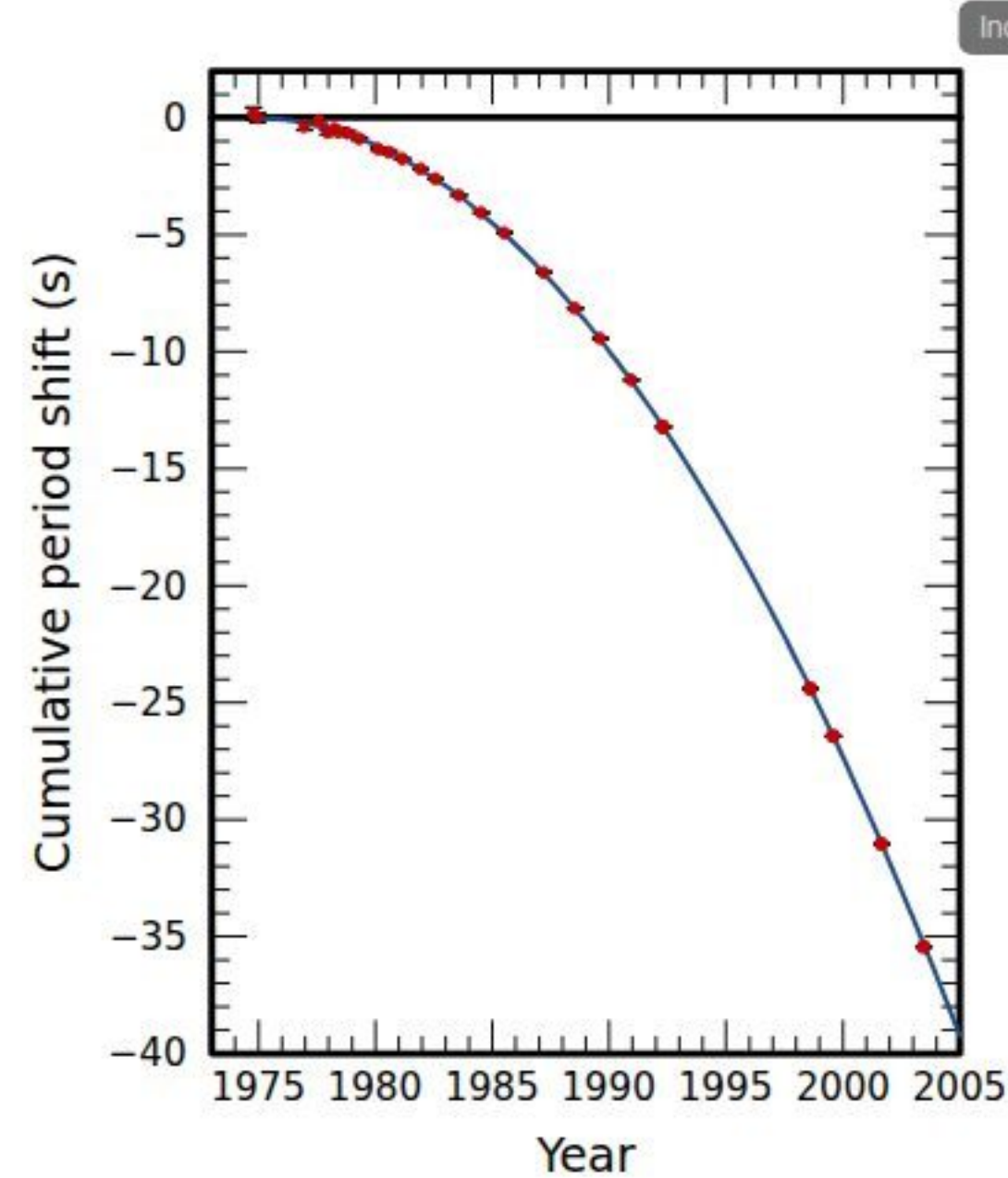
M87* (EHT, 2019) — 6.5 billion $M_{\odot}$; 42 μas ring.



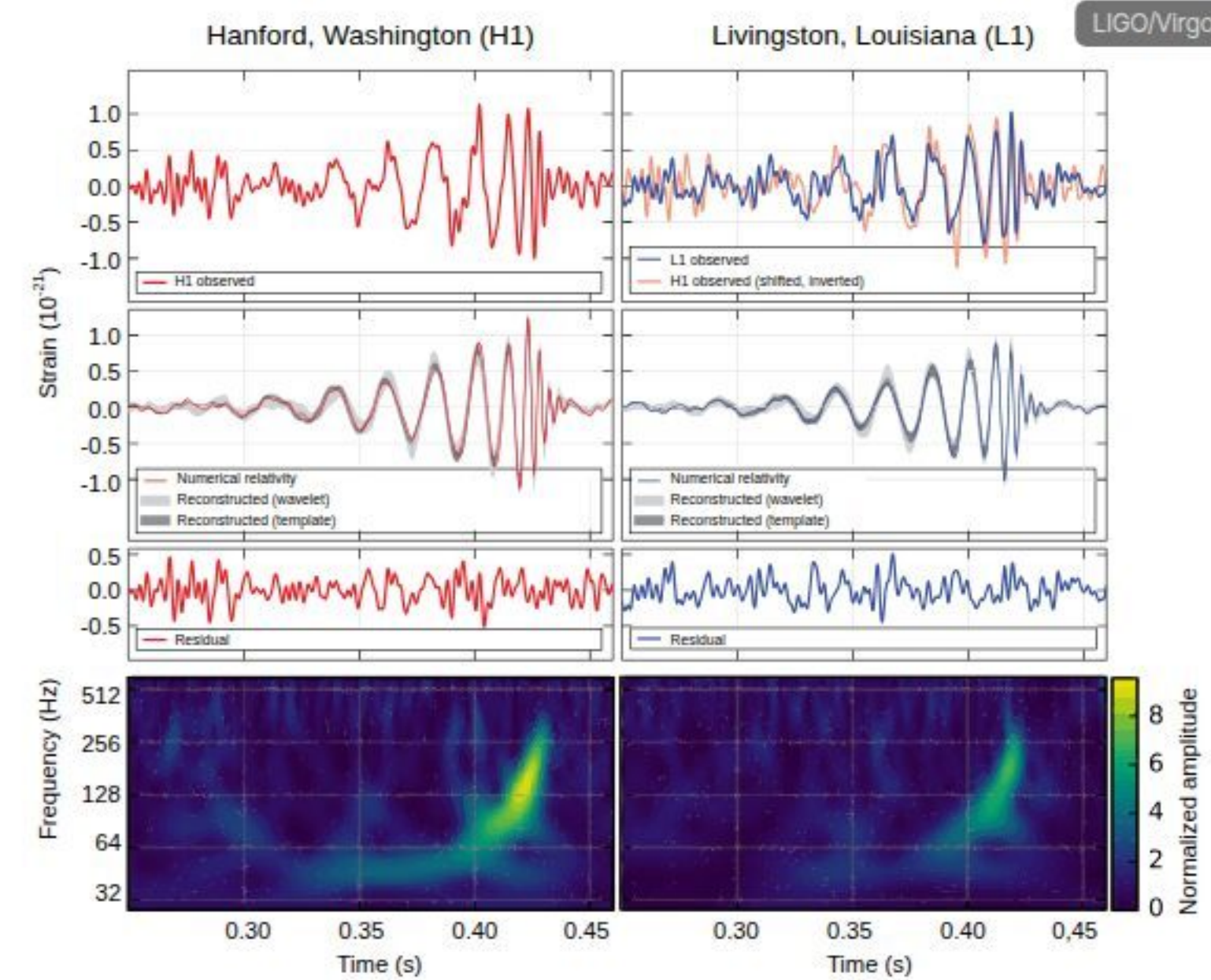
EHT Collaboration · CC BY 4.0

Sgr A* (EHT, 2022) — 4 million $M_{\odot}$, at the centre of the Milky Way.

Binary pulsars and *gravitational waves*



PSR B1513+16 (Hulse & Taylor, 1974; Nobel 1993): the orbit shrinks exactly as gravitational-wave emission predicts.



GW150914: $36 + 29 M_{\odot} \rightarrow 62 M_{\odot}$; three solar masses became gravitational waves (Nobel 2017).

The phase *along the geodesic*

$$\Phi_k = \int p_\mu^{(k)} dx^\mu$$

COVARIANT PHASE: VALID IN ANY METRIC

$$\Phi_k = \frac{m_k^2}{2E_\infty} \int \frac{dr}{\sqrt{(1 - b^2(1 - r_s/r)/r^2)}}$$

SCHWARZSCHILD, FIRST ORDER IN M^2 , ALONG THE NULL GEODESIC

One-line derivation: with E and $J = Eb$ conserved, $Et - J\phi - \int p_r dr$ cancels everything except the mass term.

1 · RADIAL: GRAVITY "HIDES"

$\Phi = \Delta m^2(r_B - r_A)/2E_\infty$, the same form as flat space. But in **local** quantities (proper distance $\int dr/\sqrt{B}$, energy E_{local}) the flat formula is wrong.

2 · NON-RADIAL: THE PATH MATTERS

The phase depends on **b**. Different paths between the same source and detector accumulate different phases.

3 · MANY PATHS: INTERFERENCE

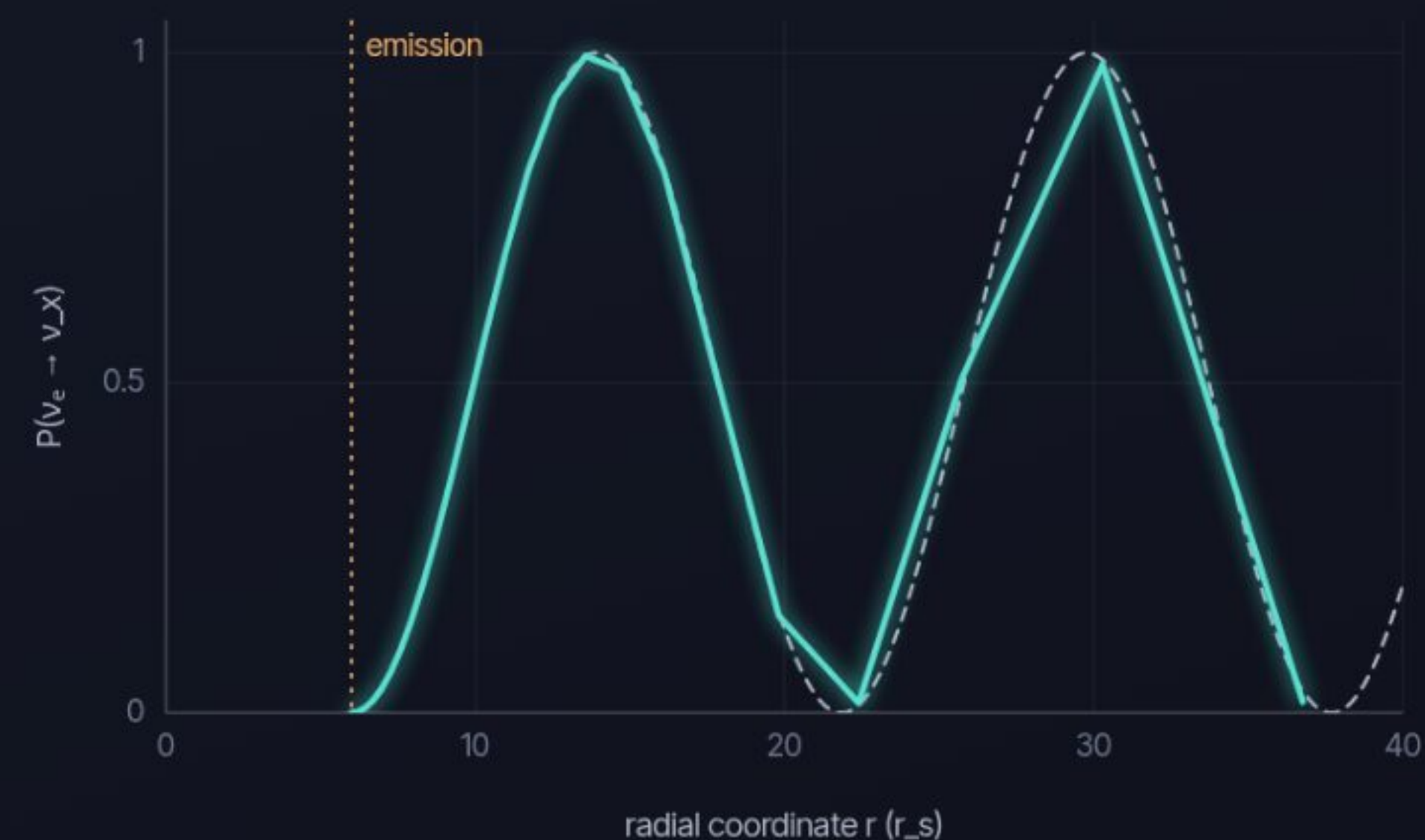
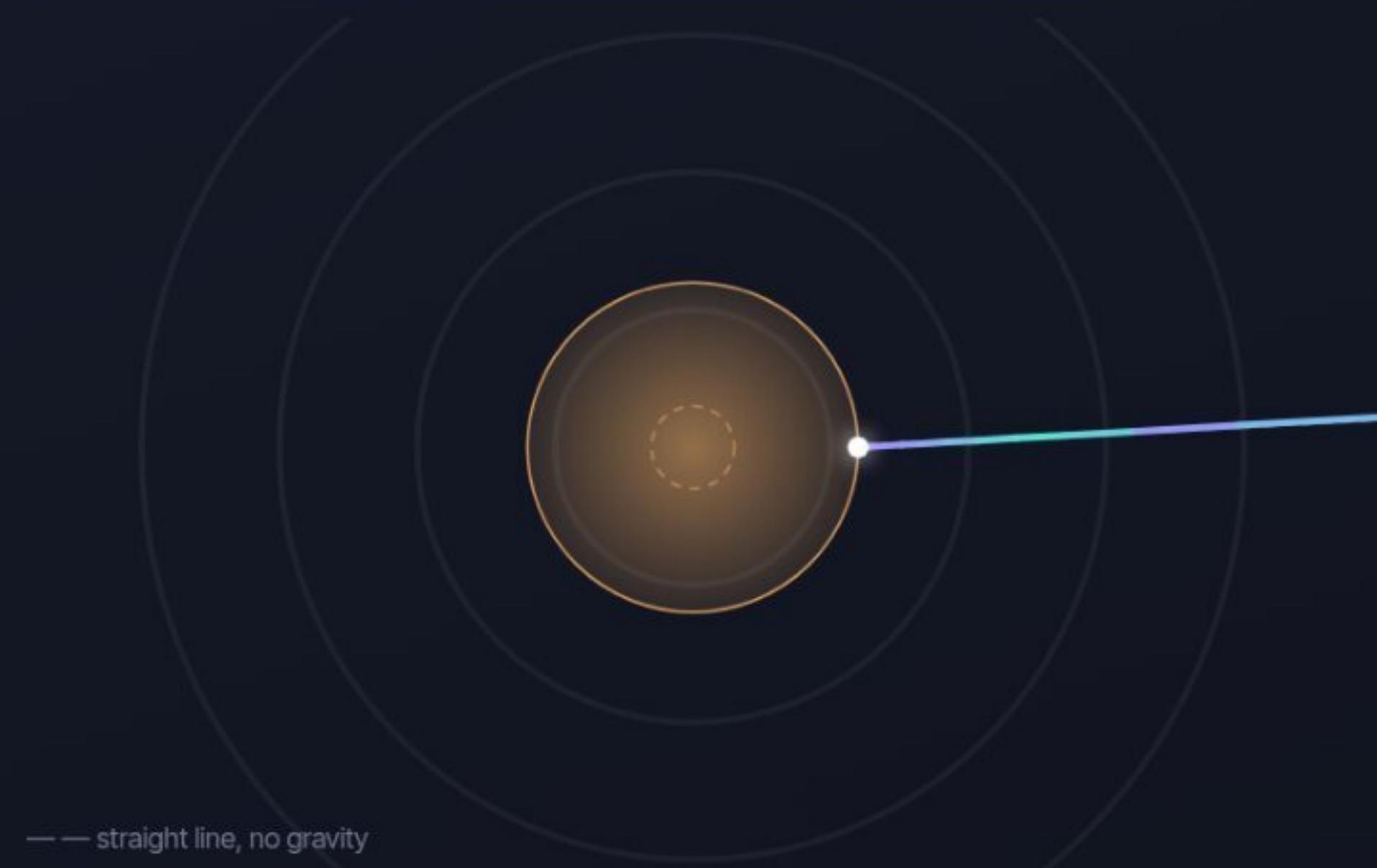
In a lens, the phase difference between paths is $\propto \mathbf{m_k}^2$, not just Δm^2 : gravity gives access to the **absolute mass**.

Redshift, bending and phase *from emission*

• AUTOMATIC · EXACT FORNENGO ET AL. INTEGRAL ALONG THE ORBIT

proto-neutron star surface

disk around a black hole



emission radius R 6.0 r_s

local emission angle α 3°

oscillation length λ 16 r_s

$E_{\text{local}}/E_\infty$ at the source 1.095

impact parameter 0.34 r_s

deflection up to 40 r_s 0.3°

phase up to $r = 40 r_s$, GR ÷ flat 0.913

II

PART TWO

Neutrinos in *curved spacetime*

They are born inside the well. They cross regions light cannot cross. And they reach us carrying a phase that remembers the path.

messengers

covariant phase

bending and redshift

lensing

DUNE

SN 1987A: neutrinos *before the light*



11

Kamiokande-II

8

IMB

5

Baksan

Within ~ 13 s, from 168,000 light-years. They arrived **hours before** the light: photons are trapped in the envelope, neutrinos are not. **99%** of the $\sim 3 \times 10^{53}$ erg of the collapse leaves as neutrinos.

A GRAVITY TEST WITH TWO DOZEN EVENTS

Neutrinos and photons suffered the same Shapiro delay in the Galaxy's potential (months) — to within $\sim 0.5\%$ (Krauss & Tremaine; Longo, 1988). Neutrinos fall like light.

The phases, and *when each one happens*

PHASE	WHEN	RADIUS	WHAT HAPPENS
collapse	-200 to 0 ms	3,000 → 30 km	the iron core loses pressure and falls at $\sim 0.25 c$
bounce	t = 0	~ 10 km	nuclear matter stiffens and the inner core rebounds
prompt shock	0 – 5 ms	10 → 100 km	born at the sonic point, it runs outward
ν_e breakout	~ 5 ms	~ 100 km	$\sim 10^{53}$ erg/s as it crosses the neutrinosphere
stalling	~ 20 ms	~ 150 km	becomes a standing accretion shock
ν heating	0.1 – 1 s	100 – 200 km	the gain region receives a few per cent of the luminosity
revival	~ 0.3 s	~ 200 km	the shock moves again: the star explodes
propagation	1 s – hours	$10^3 \rightarrow 10^6$ km	it crosses the mantle and the envelope

THE ENERGY BUDGET

The gravitational binding energy released is 3×10^{53} erg. Of that, **99%** leaves as neutrinos, and only $\sim 10^{51}$ erg becomes kinetic energy of the explosion.

A supernova is, first of all, a **neutrino event**. The light is loose change.

Before the table: a day of **silicon burning** builds the $\sim 1.4 M_{\odot}$ iron core. During infall, above 10^{12} g/cm^3 the ν_e become **trapped**, and the bounce occurs at $2.7 \times 10^{14} \text{ g/cm}^3$.

Timings from the delayed mechanism of Bethe & Wilson (1985). Radii depend on the progenitor; these are typical of $\sim 15 M_{\odot}$.

— SEEN FROM OUTSIDE

The explosion, *the remnant* and the pulsar



What the previous table describes from the inside, as a telescope would see it: the star flares, the shock breaks through the surface, the debris expands — and at the centre the neutron star remains, now a pulsar, with its wind nebula.

shock breakout

remnant

pulsar + nebula

Artist's animation · NASA's Goddard Space Flight Center
Conceptual Image Lab (animation: Jonathan North) · SVS 20413 ·
public domain. The neutrinos left hours before the first flash.

The shock *stalls*, and must be revived

The prompt shock does **not** blow up the star. It spends all its energy before getting out.

PHOTODISINTEGRATION

Breaking iron into nucleons costs **8.8 MeV per nucleon**. Crossing $0.1 M_{\odot}$ of iron consumes $\sim 1.7 \times 10^{51}$ erg, of the order of the shock's whole energy.

NEUTRINO LOSSES

As it crosses the neutrinosphere, the freshly shocked matter becomes transparent and dumps the breakout burst. More energy gone.

Within ~ 20 ms it stops at **~ 150 km**: it becomes a **standing accretion shock**.

WHAT REVIVES IT

Between the neutrinosphere and the shock lies the **gain region**, where absorption of ν_e and $\bar{\nu}_e$ beats emission. A few per cent of the luminosity deposited there is enough to push the shock back out.



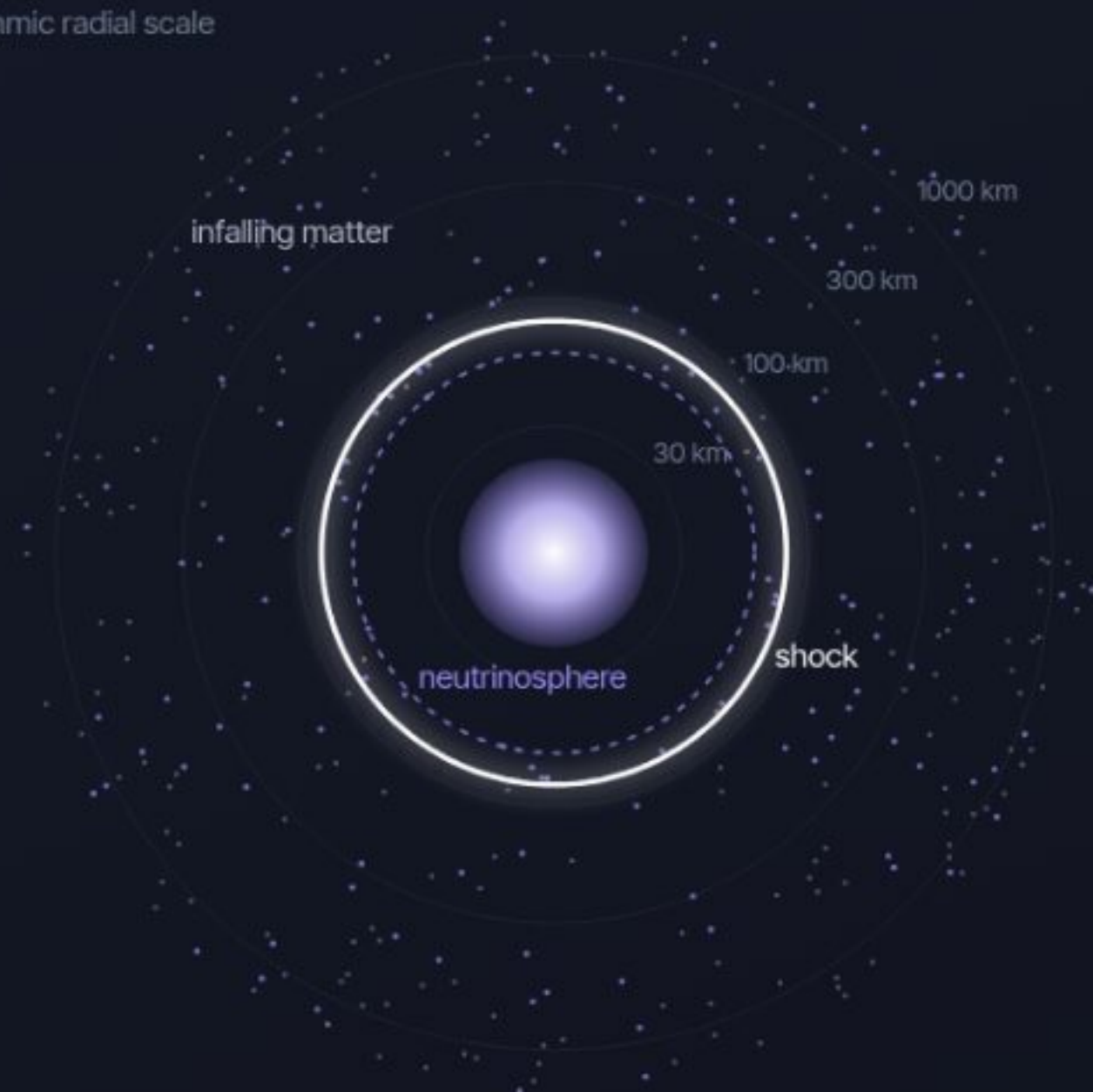
THE HEATING THAT REIGNITES THE EXPLOSION

But the deposition is marginal: in spherical symmetry it almost never suffices. What closes the budget is **convection** and the **SASI**, which hold matter longer in the gain region.

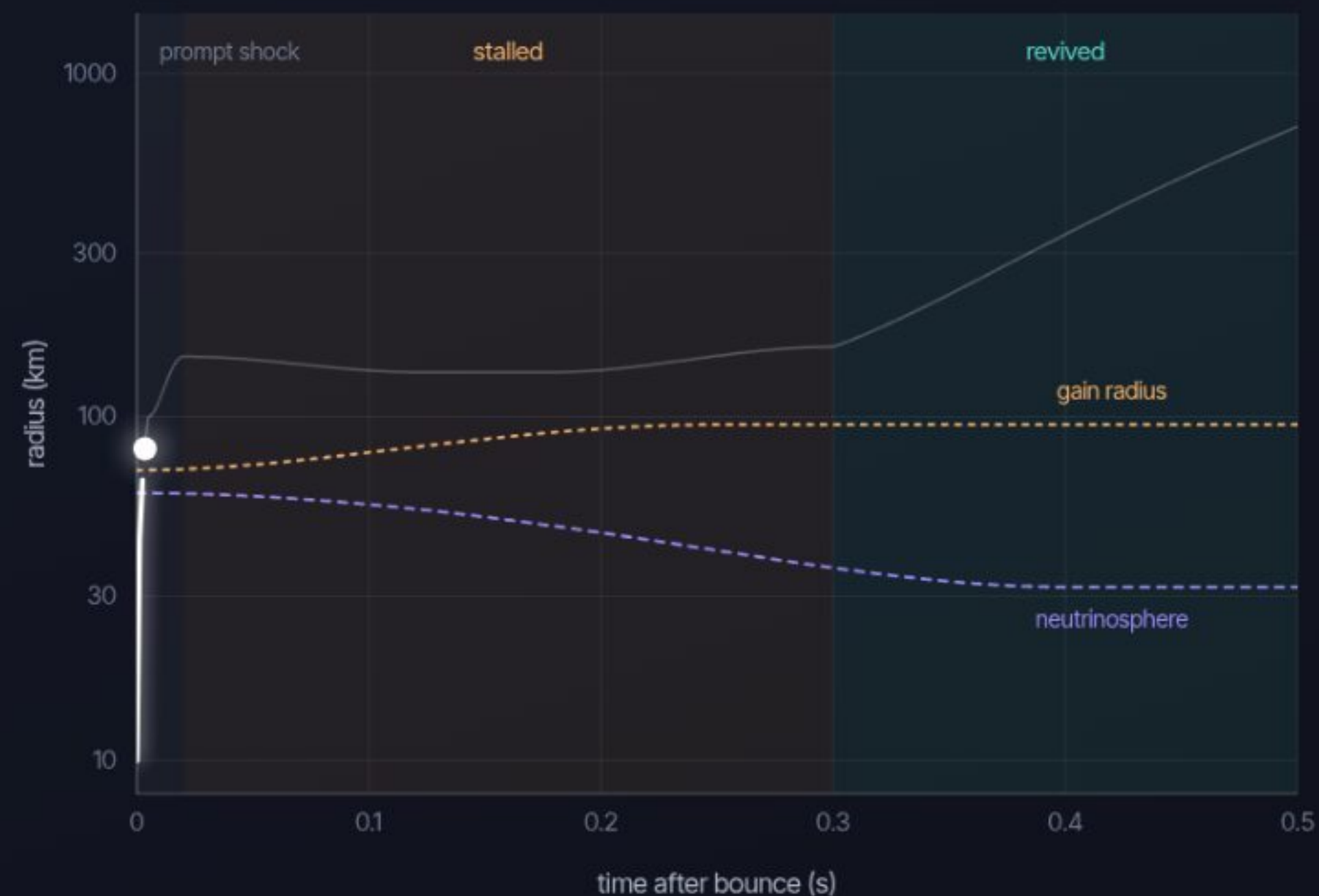
That is why the problem was only settled in 3D, and why every progenitor explodes at a different instant.

Neutrinos *relight the shock*

bounce and prompt shock
schematic · logarithmic radial scale



t = 4 ms



R_{shock} **81 km** L_{ν} **$\sim 5 \times 10^{52}$ erg/s** deposited in the gain region **$\sim 0\%$**

Schematic, not a simulation: radii and times typical of a $\sim 15 M_{\odot}$ star (those in the phase table), Bethe & Wilson (1985) mechanism. Violet: neutrinos leaving the neutrinosphere; orange: those depositing energy in the gain region.

Collapsars: when a star *fails to explode*

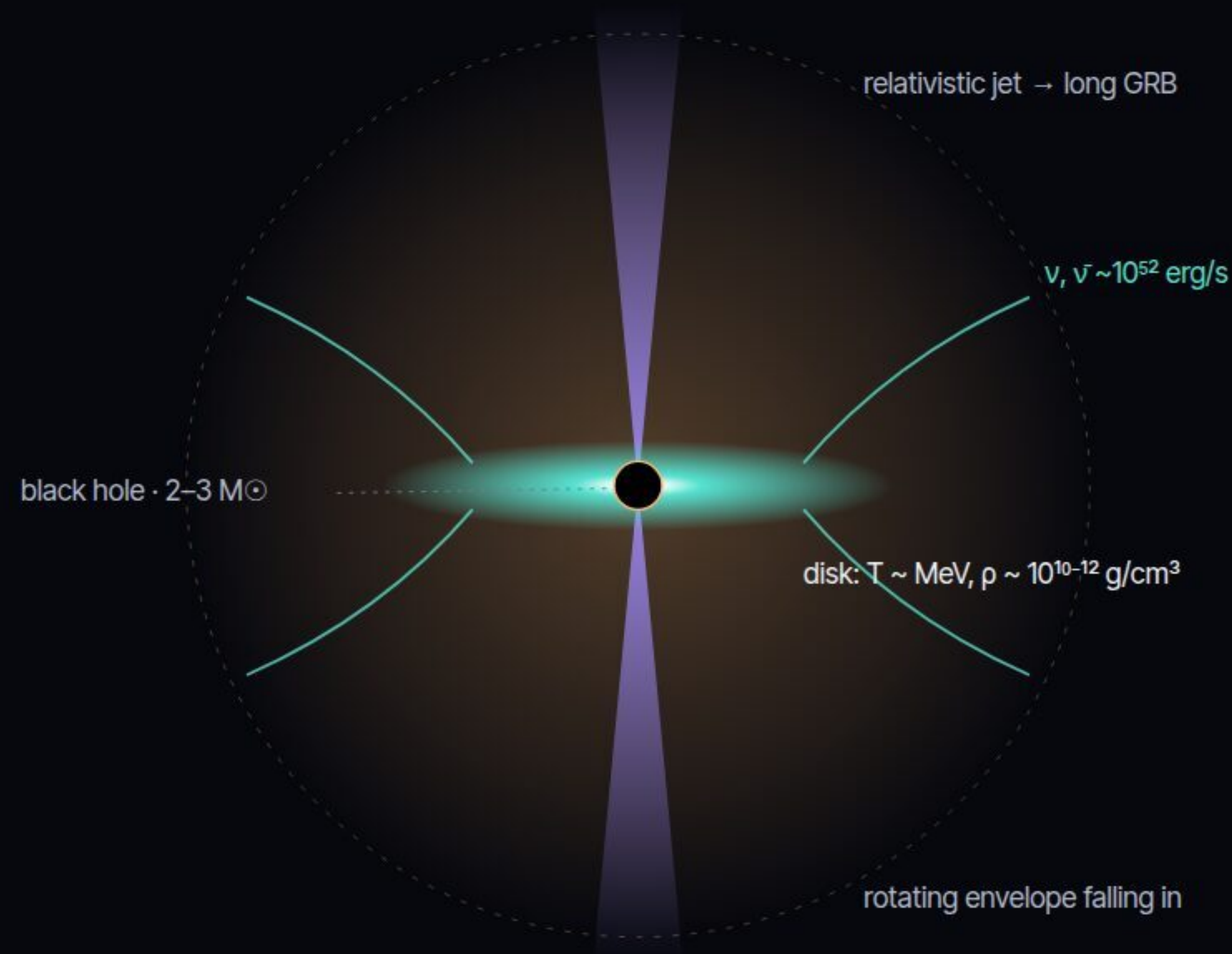


A rapidly rotating **Wolf-Rayet** star of $\sim 30 M_{\odot}$ that has lost its hydrogen envelope.

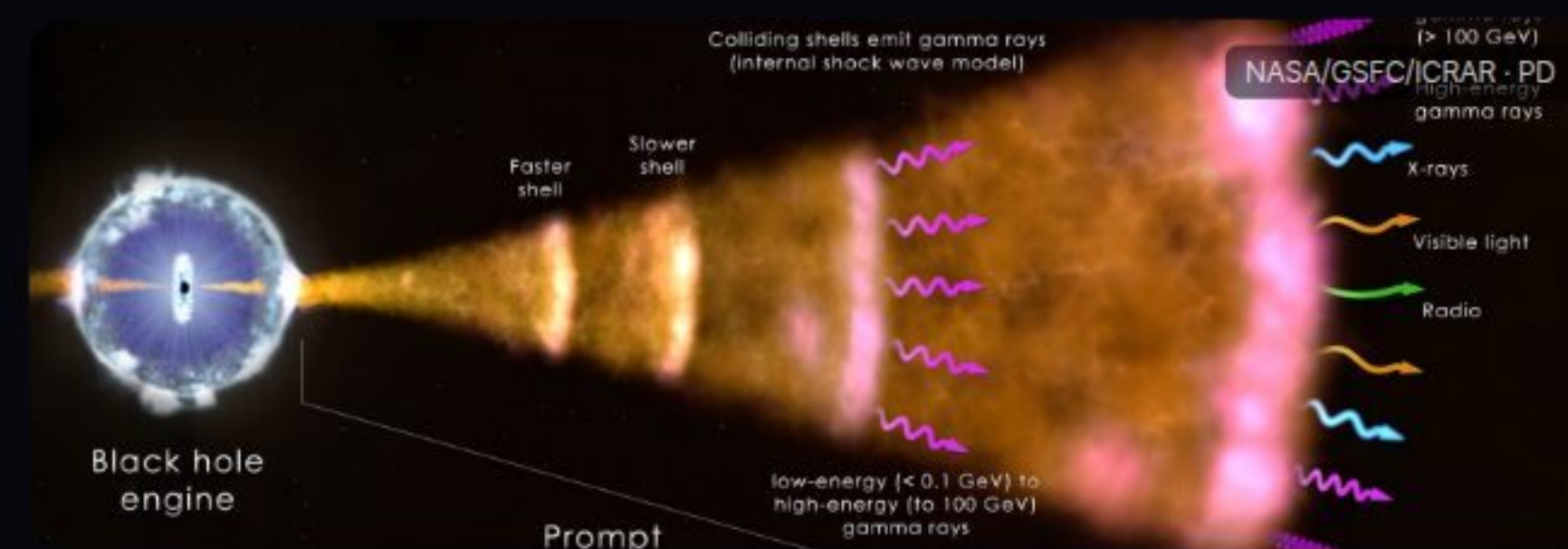
- The iron core collapses and the shock **does not revive**: a **black hole** of $\sim 2-3 M_{\odot}$ forms.
- The envelope has too much angular momentum to fall straight in: an **accretion disk** forms a few r_s out.
- Accretion at $0.01-1 M_{\odot}/s$ for tens of seconds $\rightarrow$ **relativistic jets** that drill through the star.

WR 124, imaged by JWST in 2023: the kind of star that may end this way.

An engine *cooled by neutrinos*



In the inner disk, matter is so hot and dense that **photons are trapped**. What carries the heat away are **neutrinos**: an NDAF (neutrino-dominated accretion flow).



And these neutrinos are emitted **a few r_s** from a spinning black hole, where the geometry is Kerr, not Minkowski.

Emission happens *inside the well*

SOURCE	EMISSION RADIUS	R_S/R	$E_{\text{EMITTED}}/E_\infty$
Sun, core	$\sim 0.2 R_\odot$	2×10^{-5}	1.00001
proto-neutron star, ~ 1 s	neutrinosphere ≈ 30 km	0.14	1.08
neutron star, late	≈ 12 km	0.34	1.24
collapsar disk, ISCO (Schwarzschild)	$3 r_s$, orbiting	0.33	1.41
disk around a fast Kerr hole	$< 3 r_s$	> 0.5	$> 1.4 \cdot + \text{Doppler}$

The energy DUNE measures is E_∞ . The energy that enters the MSW potential, the collisions and the collective conversions is the **local** one — up to 40% larger.

Rays leave **bent**: the neutrinosphere looks bigger, and the angular distribution of the neutrinos crossing above it changes.

In the disk the ratio is $u^t = (1 - 3M/r)^{-1/2}$: gravitational and transverse redshift together. Doppler from the orbital motion adds or subtracts depending on direction.

A gravitational lens *weighs the neutrino*



Source at 10 kpc, lens halfway. Two paths, b_+ and b_- . The common phase, $\sim m^2 L/E$, is huge and washes out with any energy resolution. The difference between paths survives:

$$\Delta\Phi_k = \frac{m_k^2 \Delta b^2}{4E} \left(\frac{1}{r_A} + \frac{1}{r_B} \right)$$

Each mass eigenstate makes **its own fringe**. Change the lightest mass: the pattern changes, and the ordering leaves a mark too.

Point lens, weak field, no matter; amplitudes $\propto \sqrt{|\mu_\pm|}$.

What gravity *can tell us*

ABSOLUTE MASS

Interference between lensed paths is sensitive to m_k^2 and to the ordering (Fornengo et al. 1997; Swami et al. 2020).

FLAVOUR NEAR THE SOURCE

MSW and collective conversions in local quantities; bent rays change the “halo” and fast flavour instabilities (Cherry et al. 2012).

BLACK-HOLE FORMATION

The neutrino signal of a supernova that turns into a black hole is **cut off abruptly**: a clock for time-of-flight and mass measurements (Beacom, Boyd & Mezzacappa 2001).

NUCLEOSYNTHESIS

In collapsar disks, the ν and $\bar{\nu}$ spectra that set Y_e are shifted and focused by GR — changing heavy-element production (Caballero, McLaughlin & Surman 2012).

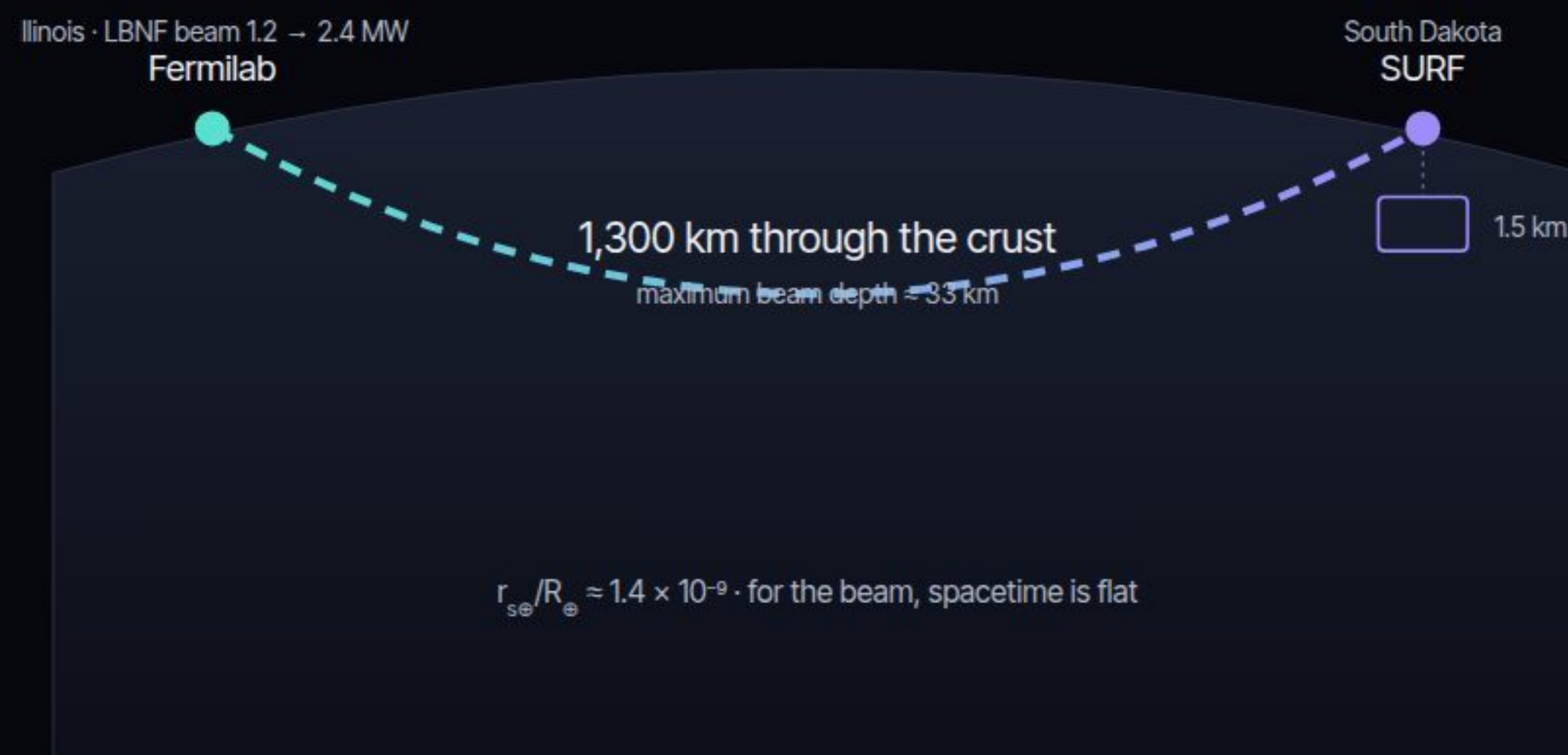
EQUIVALENCE PRINCIPLE

If different flavours felt gravity differently, oscillation would gain a term $\propto \Delta\gamma\Phi E$ (Gasperini 1988; Halprin & Leung 1991).

DECOHERENCE

Wave packets of different masses separate; in strong fields, Shapiro delays and deflections also depend on m_k .

DUNE: *1,300 km* from Fermilab to South Dakota



Far detector: 4 modules of ~17 kt of liquid argon (TPCs), 1.5 km underground. **ProtoDUNE**, at CERN, is the full-scale prototype.



Goals: δ_{CP} , **ordering**, **octant** — and supernova neutrinos.

CP and ordering *in one plot*



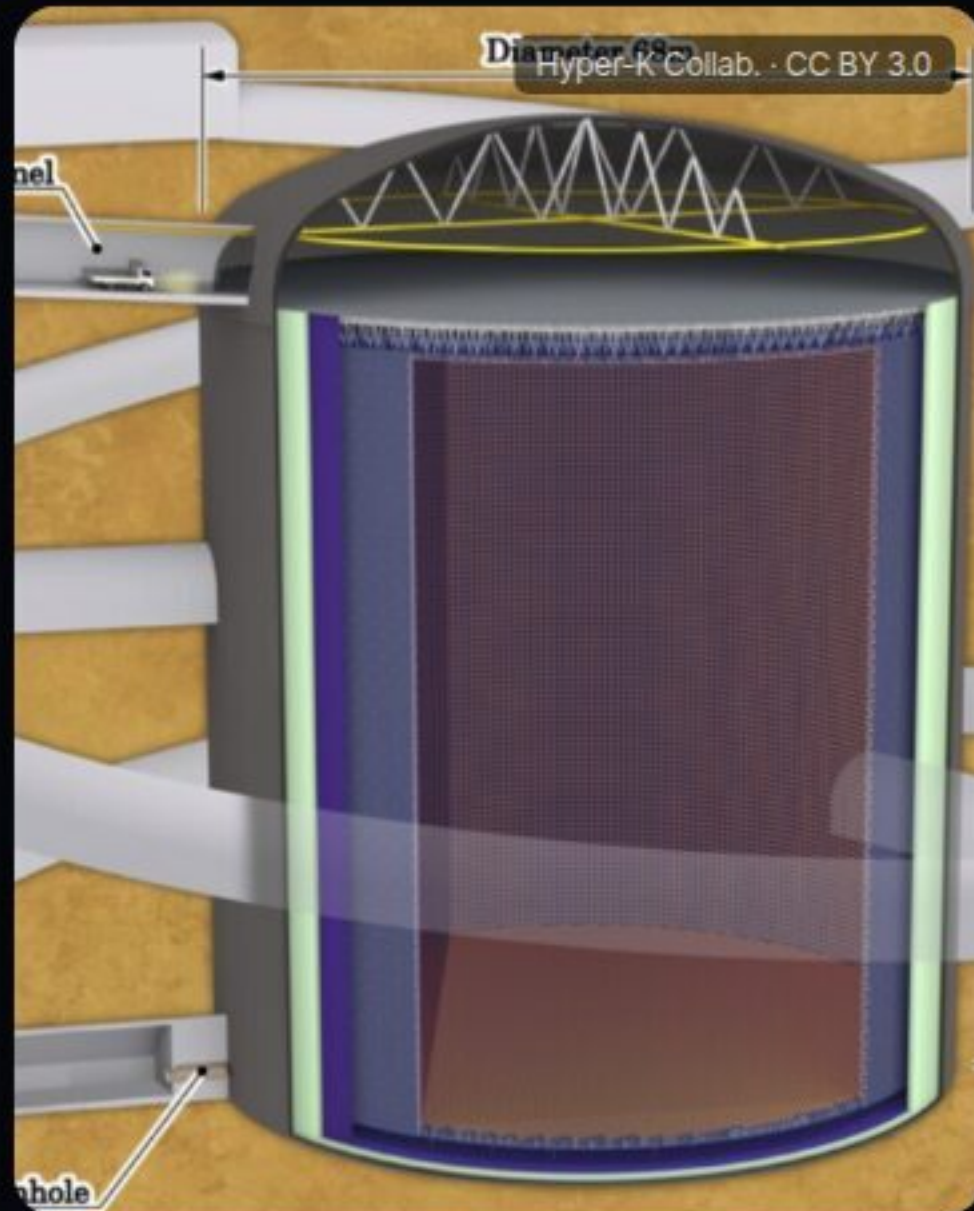
$L = 1,285$ km, crust with $\rho = 2.85$ g/cm³.
Cervera et al. (2000) expansion in θ_{13} and $\Delta m_{21}^2/\Delta m_{31}^2$.

The matter effect **separates ν from $\bar{\nu}$** depending on the ordering; δ_{CP} shifts and reshapes the peak.

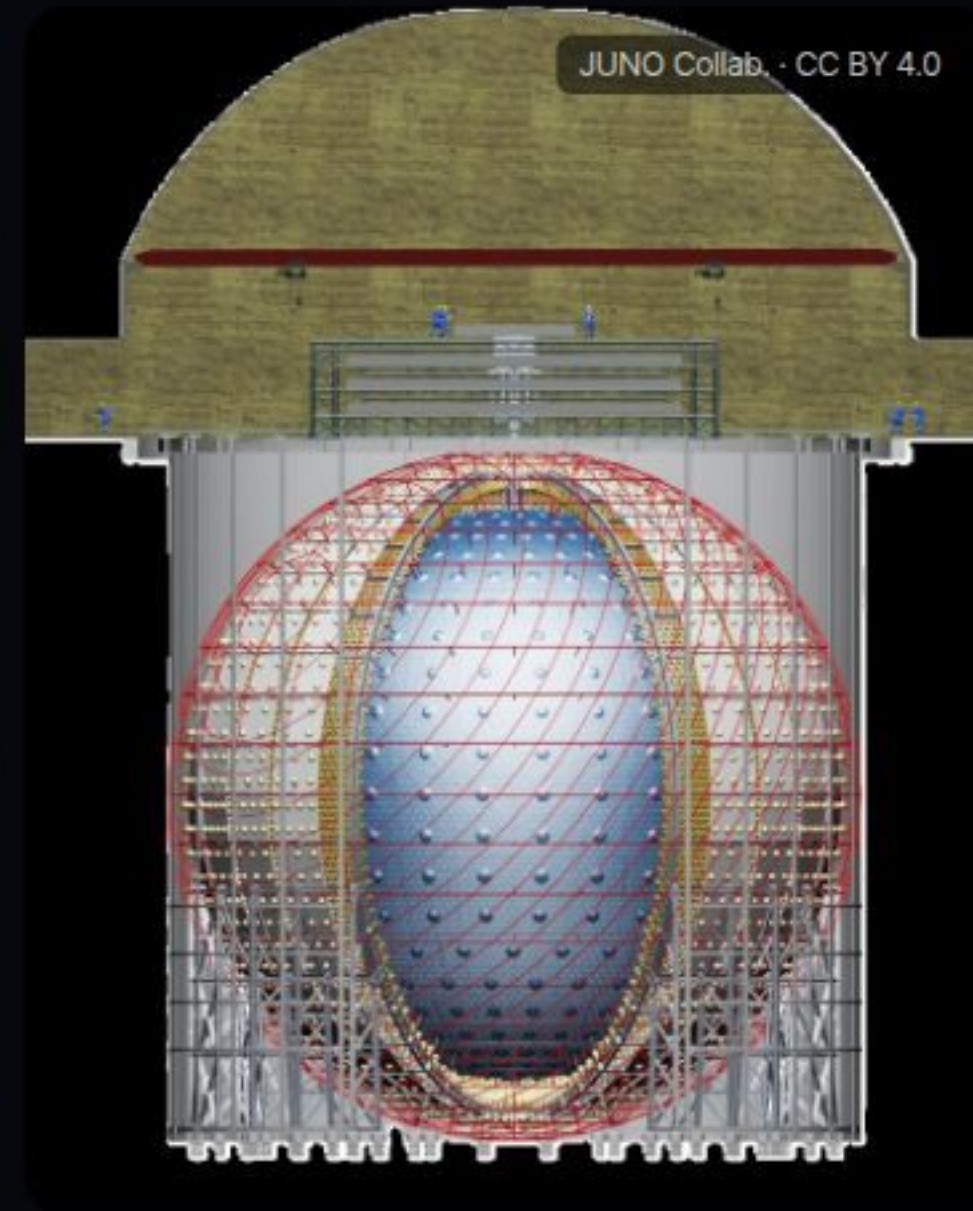
A SUPERNOVA IN THE GALAXY

DUNE is unique for ν_e ($\nu_e + {}^{40}\text{Ar} \rightarrow e^- + {}^{40}\text{K}^*$): thousands of events for an SN at 10 kpc, with the neutronization peak and, if it happens, the **cut-off** of black-hole formation.

Ears open for *the next collapse*



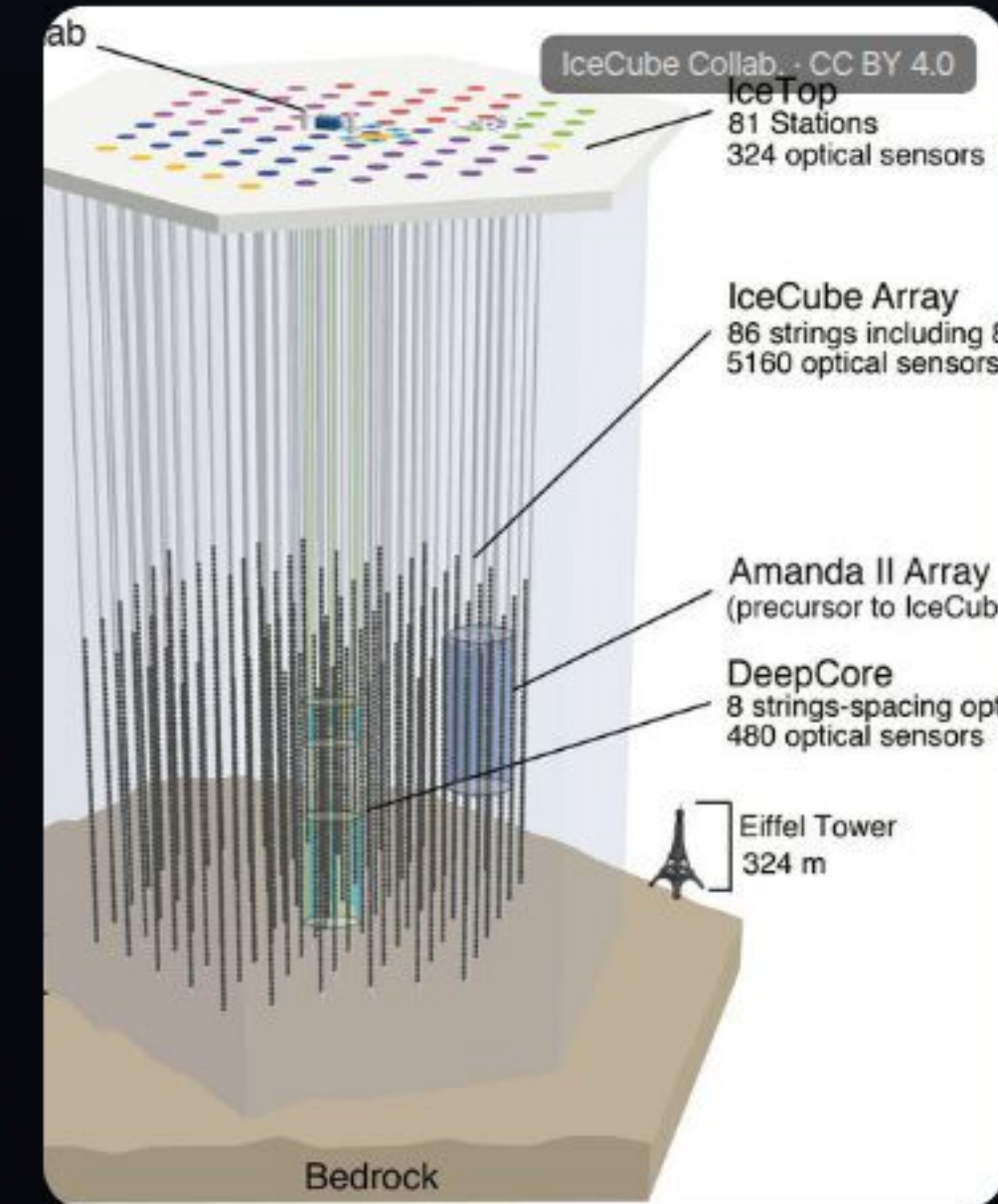
Hyper-Kamiokande — 260 kt of water, ~2028. Thousands of $\bar{\nu}_e$ from a galactic SN.



JUNO — 20 kt of scintillator in Jiangmen, running since 2025. Ordering from reactors at 53 km.



IceCube — 1 km³ of ice at the South Pole. TXS 0506+056 (2017), NGC 1068, the Galactic plane.



The detector: 5,160 optical modules between 1.45 and 2.45 km deep. And KM3NeT, in the Mediterranean, saw ~220 PeV (2025).

Two densities, *two encounters*

The star offers **every** density between 10^{14} g/cm³ in the core and almost nothing in the envelope. Each resonance picks its own.

$$\rho_{\text{res}} = \Delta m^2 \cos 2\theta \cdot m_N / (2\sqrt{2} G_F E Y_e)$$

THE SAME CONDITION, WITH EACH PAIR'S ΔM^2 AND Θ

RESONANCE	ΔM^2	P AT 20 MEV
H , θ_{13}	$2.5 \times 10^{-3} \text{ eV}^2$	$1.6 \times 10^3 \text{ g/cm}^3$
L , θ_{12}	$7.5 \times 10^{-5} \text{ eV}^2$	19 g/cm^3

Both sit in the **envelope**, not in the core: the H resonance lies tens of thousands of kilometres from the centre, and the L one further still.

ONLY ONE OF THEM MEETS THE SHOCK

The shock front takes seconds to sweep the envelope. When it passes 1.6×10^3 g/cm³ it crosses the **H** resonance. By the time it reaches the 19 g/cm³ of the **L**, it is already weak and diffuse.

That is why the rest of this talk is about H.

$Y_e = 0.5$. The density scales as $1/E$: at 5 MeV the H resonance sits at 6.3×10^3 g/cm³, and at 60 MeV at 5.3×10^2 .

The front sweeps the envelope and *crosses the H resonance*

Once revived, the shock runs outward at $\sim 1.6 \times 10^4$ km/s, sweeping the envelope.

The **MSW-H** resonance does not live at a radius: it lives at a **density**.

$$\rho_{\text{res}} \approx 1.6 \times 10^3 \text{ g/cm}^3$$

AT 20 MEV · 6.3×10^3 AT 5 MEV · 5.3×10^2 AT 60 MEV

Because the shock compresses and pushes the envelope, **the radius where that density lives moves inward** while the shock moves outward. The two are going to meet.

AFTER BOUNCE	SHOCK RADIUS	RESONANCE RADIUS
3.0 s	35,011 km	34,933 km
10.0 s	149,106 km	12,471 km

AT $T \approx 3$ S THEY COINCIDE

78 km apart out of 35,000, or **0.2%**. For a window a few seconds wide the shock front sits **right on** the resonance, and that is where the neutrino's flavour is decided.

Profiles derived from M15-7b, Garching CCSN Archive (A. Wongwathanarat), angular medians.

III

PART THREE · RESULTS

GHOST

Which density structures in a supernova change the flavour that reaches DUNE — and what it costs to ignore the only one that matters: the shock front at the MSW resonance.

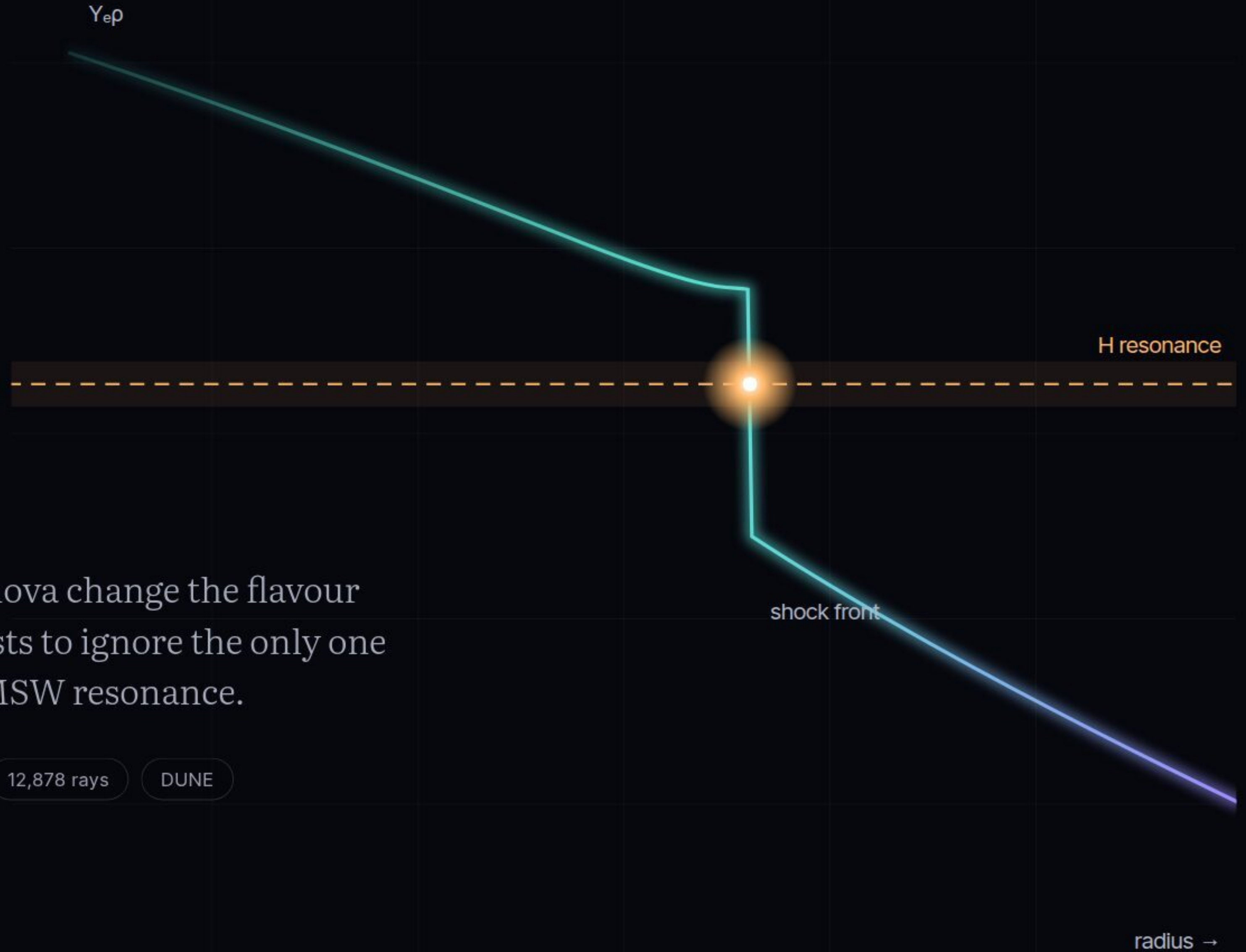
Chebyshev propagator

H resonance

shock front

12,878 rays

DUNE



GHOST: flavour oscillation *without shortcuts*

$$H(r) = \frac{1}{2E} U \text{diag}(0, \Delta m_{21}^2, \Delta m_{31}^2) U^\dagger + \text{diag}(\sqrt{2} G_F Y_e \rho(r)/m_\nu, 0, 0)$$

THREE FLAVOURS, WITH CP PHASE, ALONG EACH RAY

$$|\psi(r+\Delta r)\rangle = e^{-iH\Delta r} |\psi(r)\rangle \approx \sum_k c_k T_k(\tilde{H}) |\psi\rangle$$

CHEBYSHEV EXPANSION · RIGOROUS SPECTRAL BOUNDS · BISECTED SLICES

C core (a fork of DAEDALUS, the quantum-walks-on-graphs code by Prof. Edgard Amorim and me). The complex Hamiltonian runs in a real $2n \times 2n$ embedding; exposed as a SNEWPY *FlavorTransformation*, the rest of the pipeline is unchanged.

VALIDATION

RESULT

against NuOscProbExact (independent method)

 2.2×10^{-14} over 320 cases3 flavours + δ_{CP} in matter, against expm 7.2×10^{-15} $\rho \propto r^{-3}$ envelope: P_{ee} against $\sin^2 \theta_{13}$ 0.0223635 vs
0.0223608DUNE spectrum, 200 energies $\times 2$

13 ms

FROM PART II TO HERE

Along a radial path, the gravitational terms in the phase **cancel** (Fornengo et al.). What decides the flavour leaving the star is the **density structure** the neutrino crosses.

Why *Chebyshev*

$$|\psi(x+\Delta x)\rangle = e^{-iH\Delta x} |\psi(x)\rangle$$

THE TASK: APPLY THE EXPONENTIAL, SLICE BY SLICE OF THE PROFILE

$$e^{-iH\Delta x} \approx \sum_k c_k T_k(\tilde{H}), \quad c_k \propto (-i)^k J_k(\lambda\Delta x)$$

Chebyshev polynomials in H ; COEFFICIENTS = BESSEL FUNCTIONS



1 · GUARANTEED PRECISION

The coefficients fall super-exponentially once the order passes $\sim \lambda\Delta x$ (λ : spectral radius). With rigorous spectral bounds, GHOST knows how many terms reach machine precision — hence the 2.2×10^{-14} against NuOscProbExact.

2 · LARGE STEPS

Each constant-density slice is propagated **in one go**, even across many L_{osc} . A Runge–Kutta would need steps shorter than L_{osc} and would accumulate error and loss of unitarity over thousands of km.

3 · NO DIAGONALIZATION

Only matrix $\times$ vector products. The norm is conserved to round-off.

4 · ANY HERMITIAN H

The same code does 2 or 3 flavours, CP phase, steriles and, via quantum trajectories, decoherence. Mass and matter enter as “self-loops” on the diagonal; the complex H , through a real $2n \times 2n$ embedding.

Six *blue* supergiants, exploded in 3D

Models by A. Wongwathanarat and H.-Th. Janka built for **SN 1987A**, whose progenitor, **Sanduleak -69 202**, was a **blue** supergiant — the atypical case that made it famous.

MODEL	PROGENITOR	REFERENCE
M15-7b	binary merger (Menon & Heger)	Utrobin+ 2021
W16-3	single star	Utrobin+ 2019
W18-3	single star	Utrobin+ 2019
W18r-2	single star	Utrobin+ 2019
W18x-2	single star	Utrobin+ 2019
W20-5	single star	Utrobin+ 2019

Janka advised leaving out B15-1 and N20-4 (envelope attached by hand to a helium core). M15-7b reproduces Sanduleak's colour and luminosity and the SN 1987A light curve.

WHAT THEY SIMULATE

Neutrino-driven explosion in 3D (PROMETHEUS-HOTB): from bounce, through shock revival, to long after breakout at the surface — with element mixing and ^{56}Ni . Light curves (CRAB) are compared with SN 1987A.

WHAT WE RECEIVED

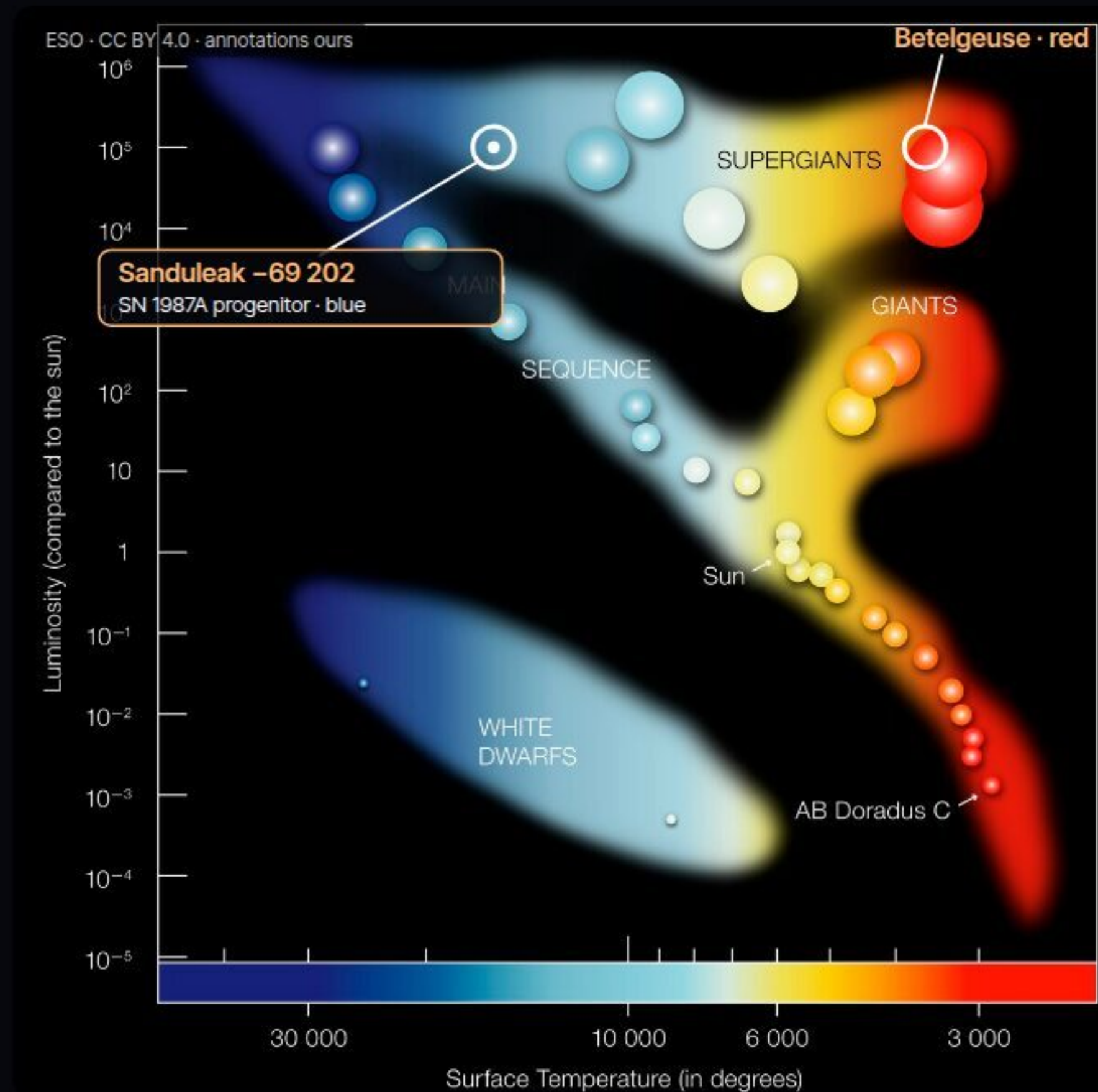
3D snapshots on a Yin-Yang grid: $2 \times 137 \times 47$ directions $\times$ 2400 radial zones = **12,878 rays**, with density, Y_e , temperature, velocity and composition. Epochs from ~ 0.3 s to ~ 10 s after bounce — the window in which the shock crosses the H resonance.

BLUE, NOT RED — AND WHY IT MATTERS

The next Galactic supernova will almost certainly be a **red** supergiant. The (t, E) band where the effect lives sits inside the helium core, which does not know the colour of the envelope; Janka expects the result to carry over. It is an argument, not yet a calculation.

Data: Garching CCSN Archive, courtesy of H.-Th. Janka and A. Wongwathanarat (MPA). Wongwathanarat, Müller & Janka, A&A 577, A48 (2015); Utrobin et al., A&A 624, A116 (2019); ApJ 914, 4 (2021).

Blue and red supergiants



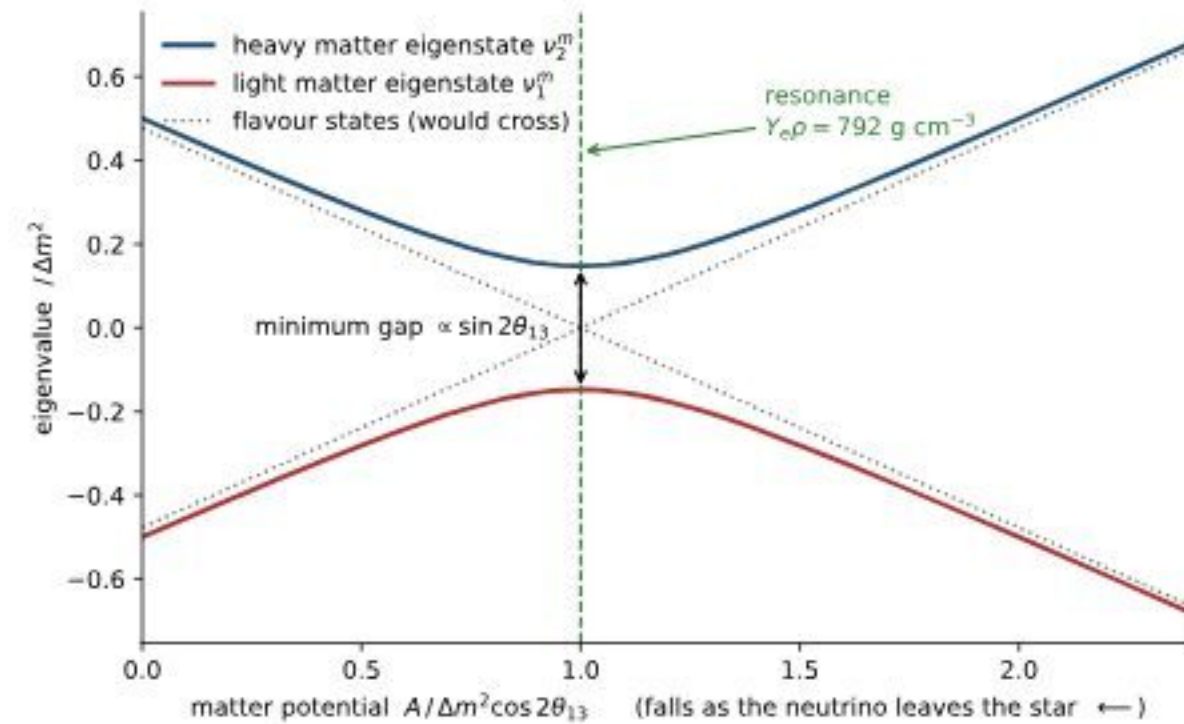
Stars above $\sim 8 M_{\odot}$ end their lives at the top of the diagram, as **supergiants**. The colour tells the size of the hydrogen envelope:

	BLUE	RED
example	Sanduleak -69 202	Betelgeuse
temperature	$\sim 16\,000\text{ K}$	$\sim 3\,600\text{ K}$
radius	$\sim 50\text{ R}_{\odot}$	$\sim 800\text{ R}_{\odot}$
luminosity	$\sim 10^5\text{ L}_{\odot}$	$\sim 10^5\text{ L}_{\odot}$

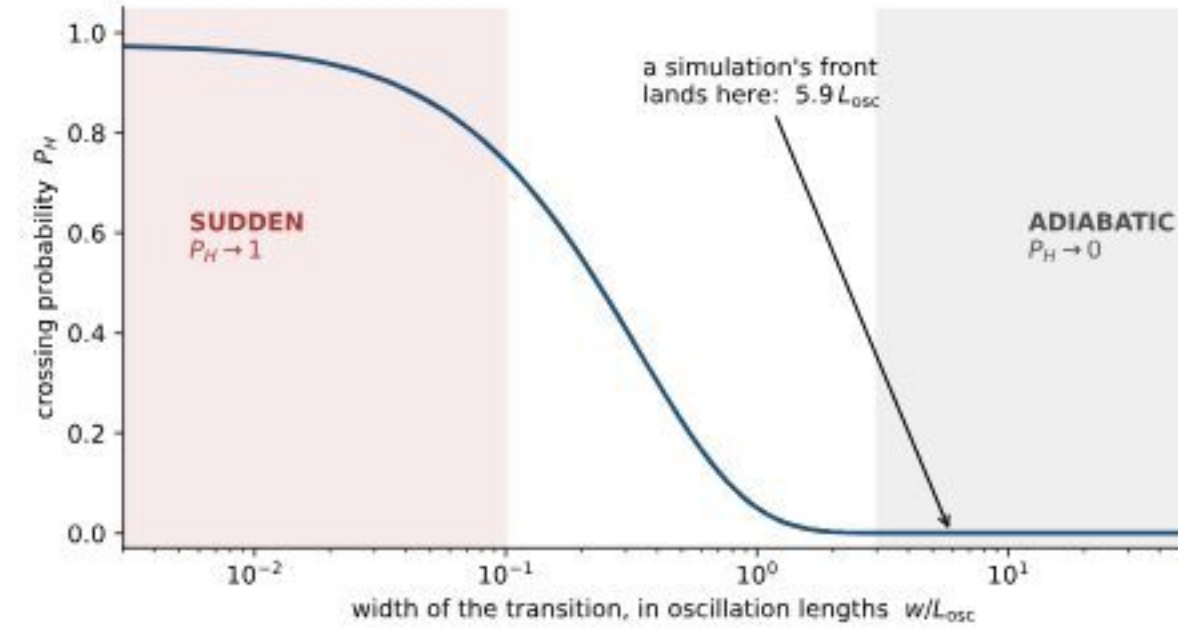
SN 1987A was the exception: a compact **blue** supergiant. The six Garching models were built for it. Most core-collapse supernovae — and probably the next one in the Galaxy — come from **red** supergiants.

For our effect what matters is the helium core, which is similar in both: the envelope lies far beyond the H resonance.

A shock wave crosses *the resonance*



At the H resonance the two matter eigenstates almost touch.



Cross slowly: $P_H = 0$. Cross a discontinuity: $P_H \rightarrow 1$.

For the first seconds after bounce, **the shock front sits on the H resonance**. What decides the flavour DUNE measures is the shape of a shock wave.

ORDERING	ADIABATIC P_{EE}
normal	0.022
inverted	0.30

A factor of **13** in ν_e survival — and argon counts precisely ν_e .

Three steps, and *only two have matured*

1 · HYDRODYNAMICS

3D models deliver time- and angle-resolved fluxes.

2 · FLAVOUR TRANSFORMATION — THIS STEP

In SNEWPY 1.5.1, for standard MSW, only two numbers: $P_H = 0$ and $P_H = 1$. None of the ten classes depends on post-bounce time.

3 · DETECTOR

SNOWGLOBES turns fluxes into event rates.

“the shock front is steepened artificially to reintroduce the physical requirement of a density discontinuity which is lost in hydrodynamic simulations”

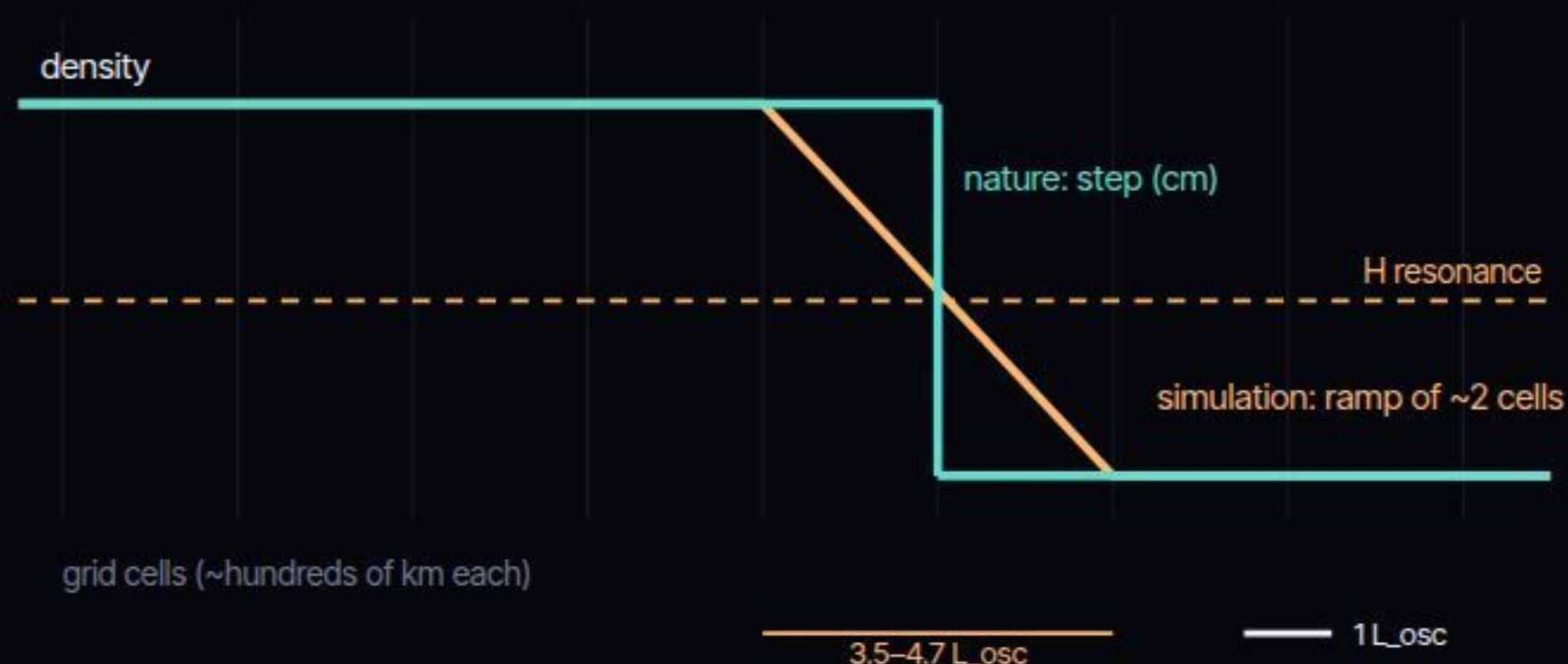
Barger, Huber & Marfatia, 2005

Tomàs et al. (2004) already steepened the front by hand. Twenty years later the correction has lapsed: recent work passes 24 3D models through an adiabatic constant.

Steepening the front: *restoring the step*

*“the shock front is **steepened artificially** to reintroduce the physical requirement of a **density discontinuity** which is lost in hydrodynamic simulations”*

Barger, Huber & Marfatia, 2005



1 · IN NATURE: A STEP

Density jumps over centimetres. For a neutrino oscillating over tens of km, that is a **perfect discontinuity**.

2 · IN THE SIMULATION: A RAMP

Shock capturing: numerical viscosity spreads the shock over ~1.6–1.7 cells. Position and strength right; **width set by the grid**, not by physics.

3 · HYDRO DOESN'T CARE; THE NEUTRINO DOES

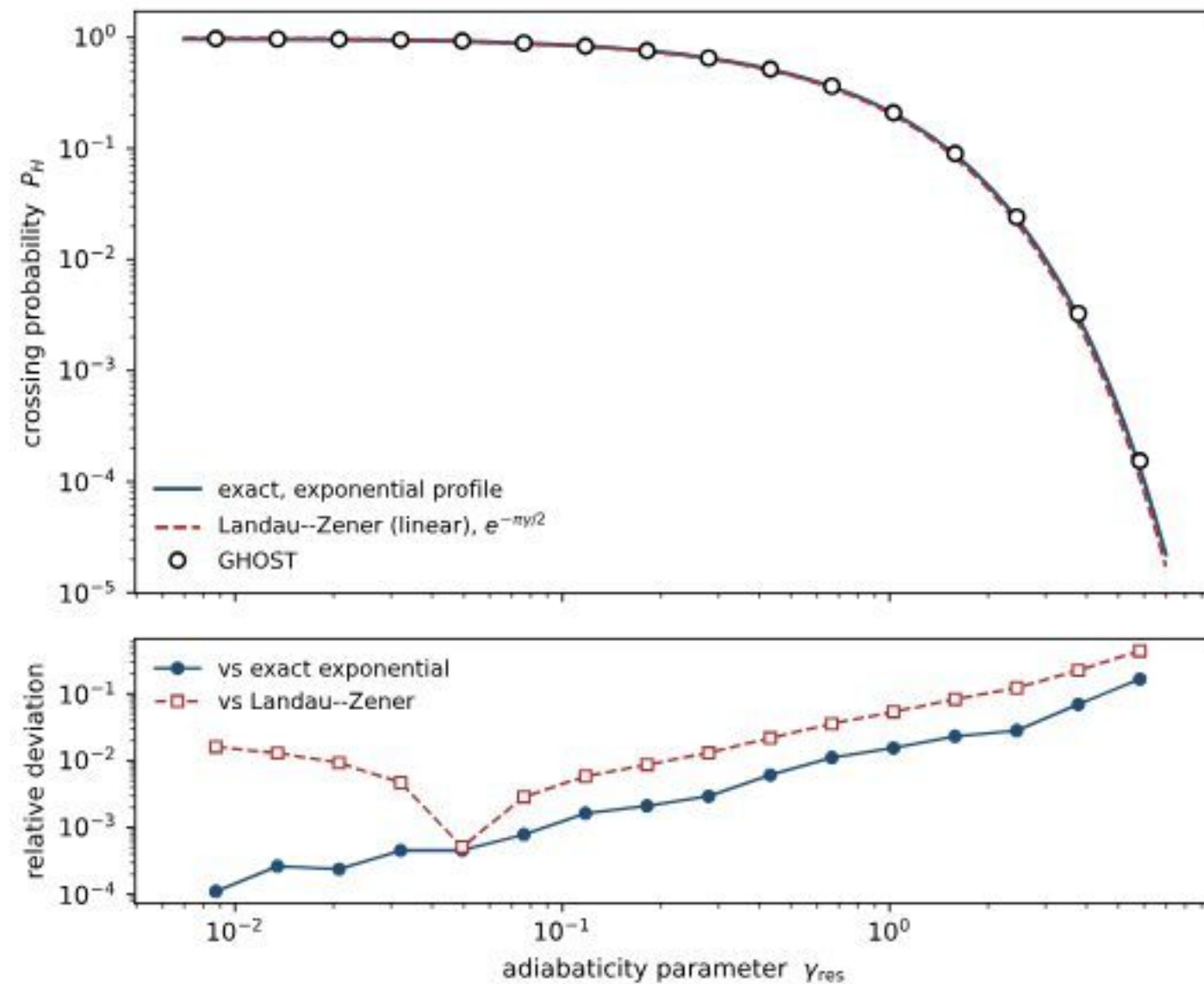
Conservation across the shock does not depend on the width. MSW depends only on it: $\gamma \propto \text{width} / L_{\text{osc}}$. Ramp: $\gamma \approx 5$, $P_H \approx 0$. Step: $\gamma \approx 0.01$, P_H up to ~0.85.

4 · THE FIX: STEEPEN BY HAND

Locate the front and replace the ramp with a step, keeping the densities on both sides. “Artificial” restores a **physical** property the grid destroyed.

The simulation gets where the shock is right — but not how abrupt it is. And the neutrino only cares about that.

Adiabaticity, and *the length that measures everything*



P_H against the adiabaticity parameter: **GHOST** (circles) on top of the exact exponential profile and of the Landau-Zener formula. Below, the relative deviation from each.

At the resonance the two matter eigenstates nearly touch. If the neutrino crosses **slowly**, it follows the eigenstate and $P_H = 0$. If it crosses **fast**, it hops, and $P_H \rightarrow 1$.

“Slow” and “fast” are measured against a single length, the oscillation length **at** the resonance:

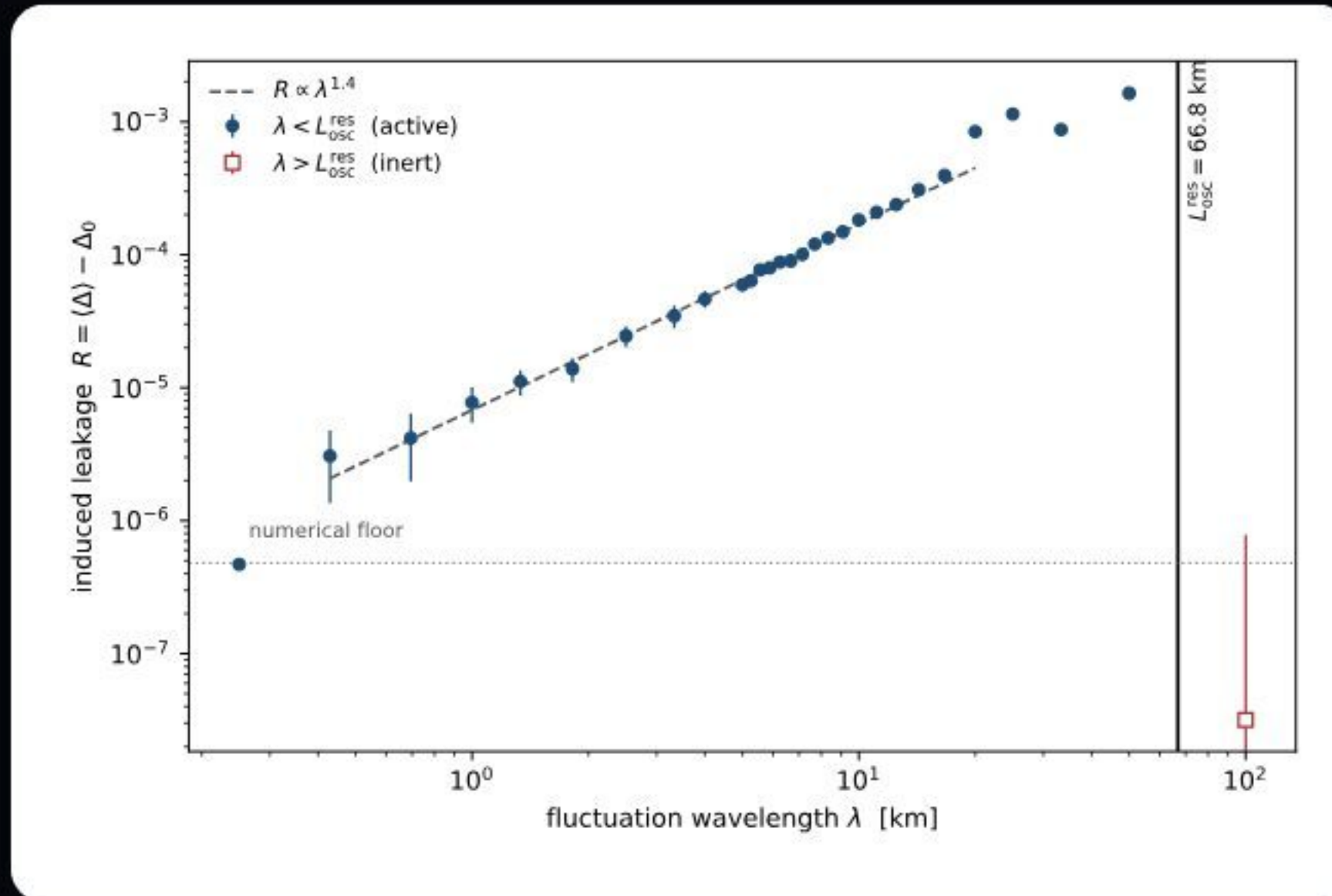
$$L_{\text{osc}} = 4\pi E / (\Delta m^2 \sin 2\theta_{13})$$

AT THE H RESONANCE, AT 20 MEV: **67.6 KM**

THIS IS THE RULER FOR THE NEXT SLIDES

The criterion that a structure only perturbs the conversion if it is **shorter than L_{osc}** is measured in this unit. And the front width recorded in M15-7b, **284 km**, is exactly **4.2 L_{osc}** .

What the resonance *can see*



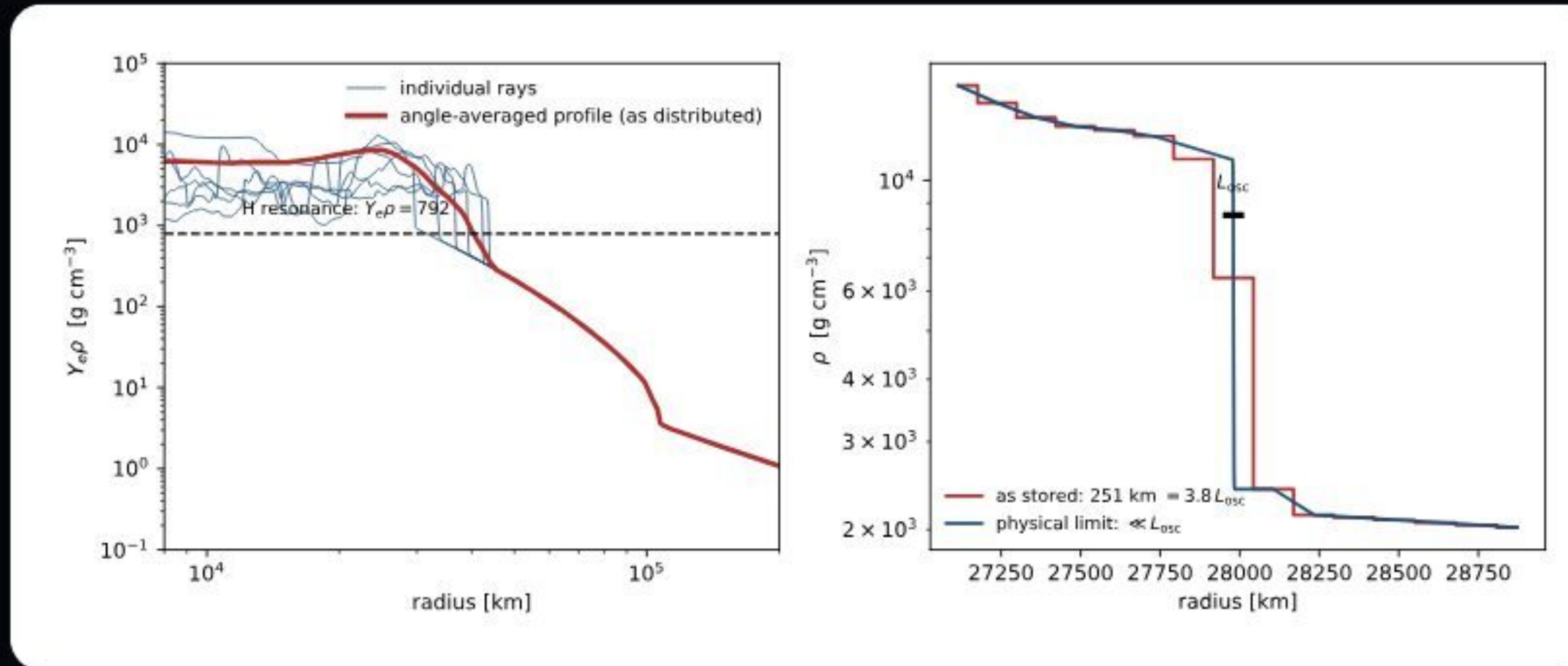
Induced leakage against fluctuation wavelength: the response dies above L_{osc} at the resonance.

A structure perturbs the conversion only if it is **shorter than L_{osc}** at the resonance. Below that, the effect grows as the **square** of the contrast — measured as the 2.02 ± 0.03 power — and saturates at 8%.

STRUCTURE	CONTRAST	SCALE < L_{osc} ?	VERDICT
turbulence	7×10^{-4}	✓	111× below saturation
convection, SASI	$\mathcal{O}(1)$	✗ far too large	nothing
shock front	$\mathcal{O}(1)$	✓ discontinuity	saturated

— BUT IN A SIMULATION...

The front *has no physical width*



Each ray crosses the resonance sharply; the angular average does not. Right: the front as stored and at the physical limit.

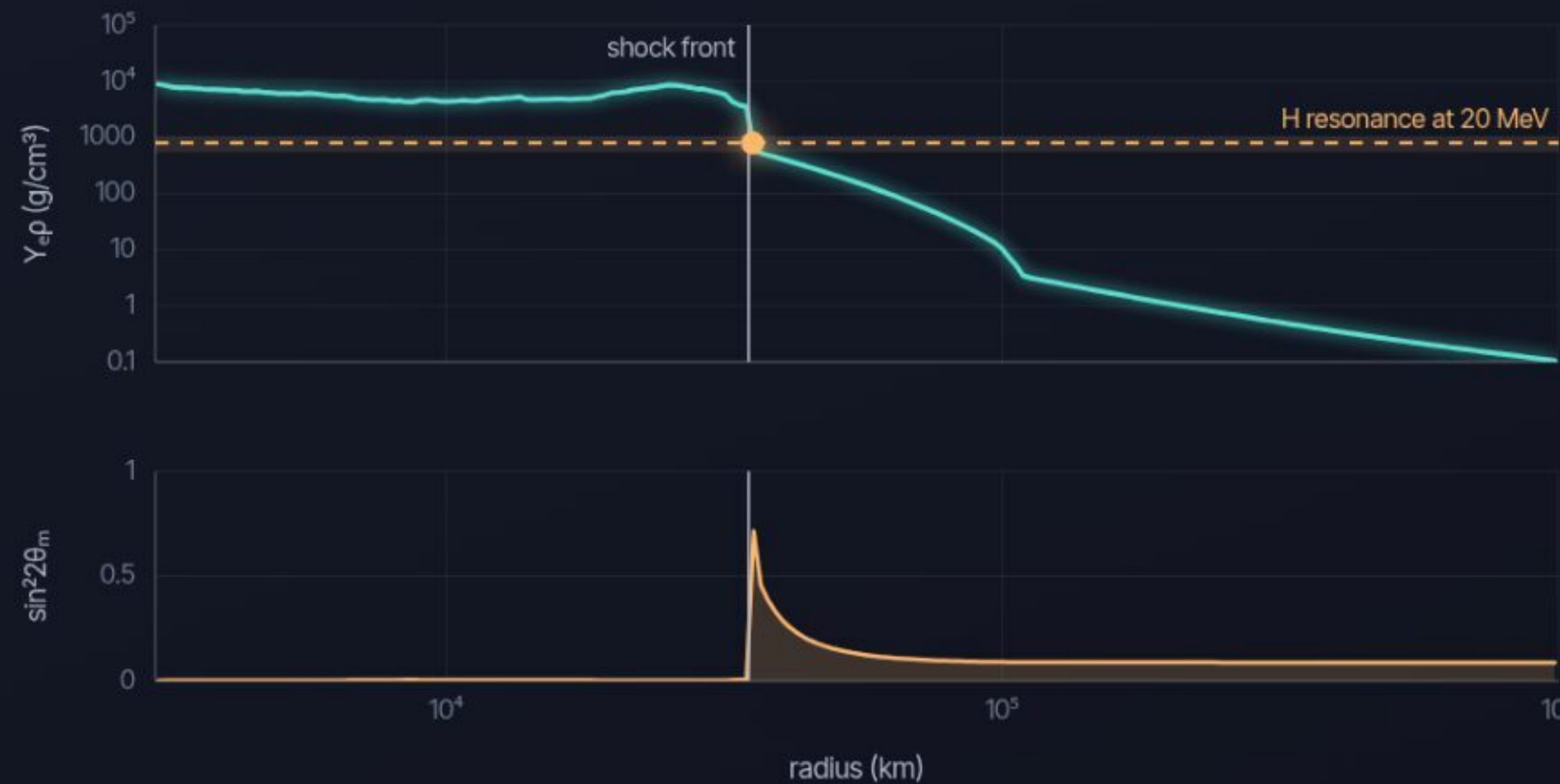
The real shock is centimetres wide. The file stores the width of the **numerical scheme**.

MODEL (GARCHING)	WIDTH (L_{osc})	ZONES
M15-7b	4.20	1.67
W16-3	3.81	1.65
W18-3	3.52	1.64
W18r-2	3.76	1.63
W18x-2	4.67	1.65
W20-5	4.22	1.65

All above the $\approx 3 L_{\text{osc}}$ threshold: propagating the file as it comes returns the adiabatic result. **A null produced by resolution.**

The shock overtakes the resonance *at $t \approx 3$ s*

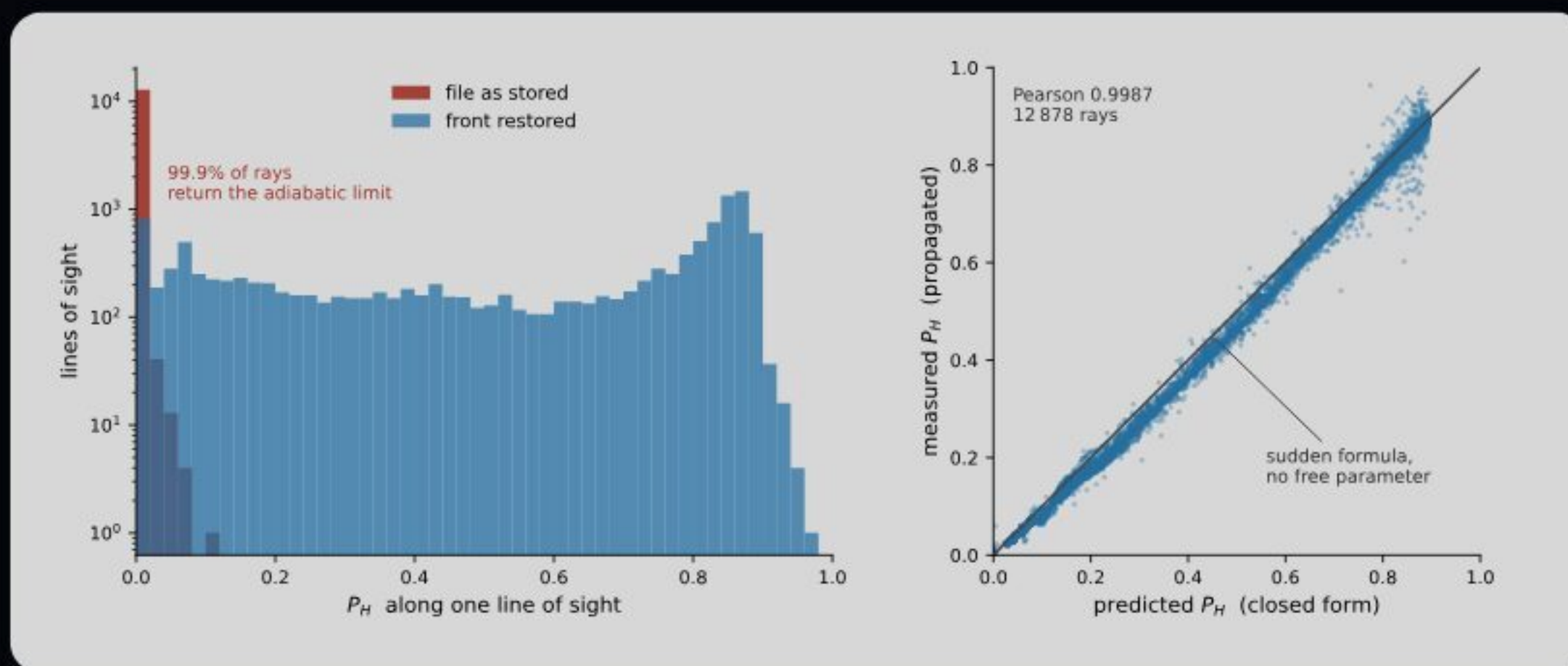
• INTERACTIVE · M15-7B



time after bounce 3.00 s energy 20 MeV front width w 4.17 L_{osc}

R_{shock} 35.0×10^3 km R_{res} 35.6×10^3 km width 278 km P_H $3.0e-4$ P_{ee} (NO) 0.022

Restoring the front removes *half of P_H*



The whole sky of M15-7b: **12,878 rays**, each propagated twice.

0.44–0.76

absolute ΔP_H , six progenitors, 5 to 100 MeV (median 0.56)

The file as stored returns the adiabatic limit in **99.9%** of rays. With the front restored, the spread (s.d. 0.32) is where one supernova is drawn from.

The closed form tracks each ray: **Pearson 0.9987**, no free parameter.

A few per cent, *with a sign set by the source*

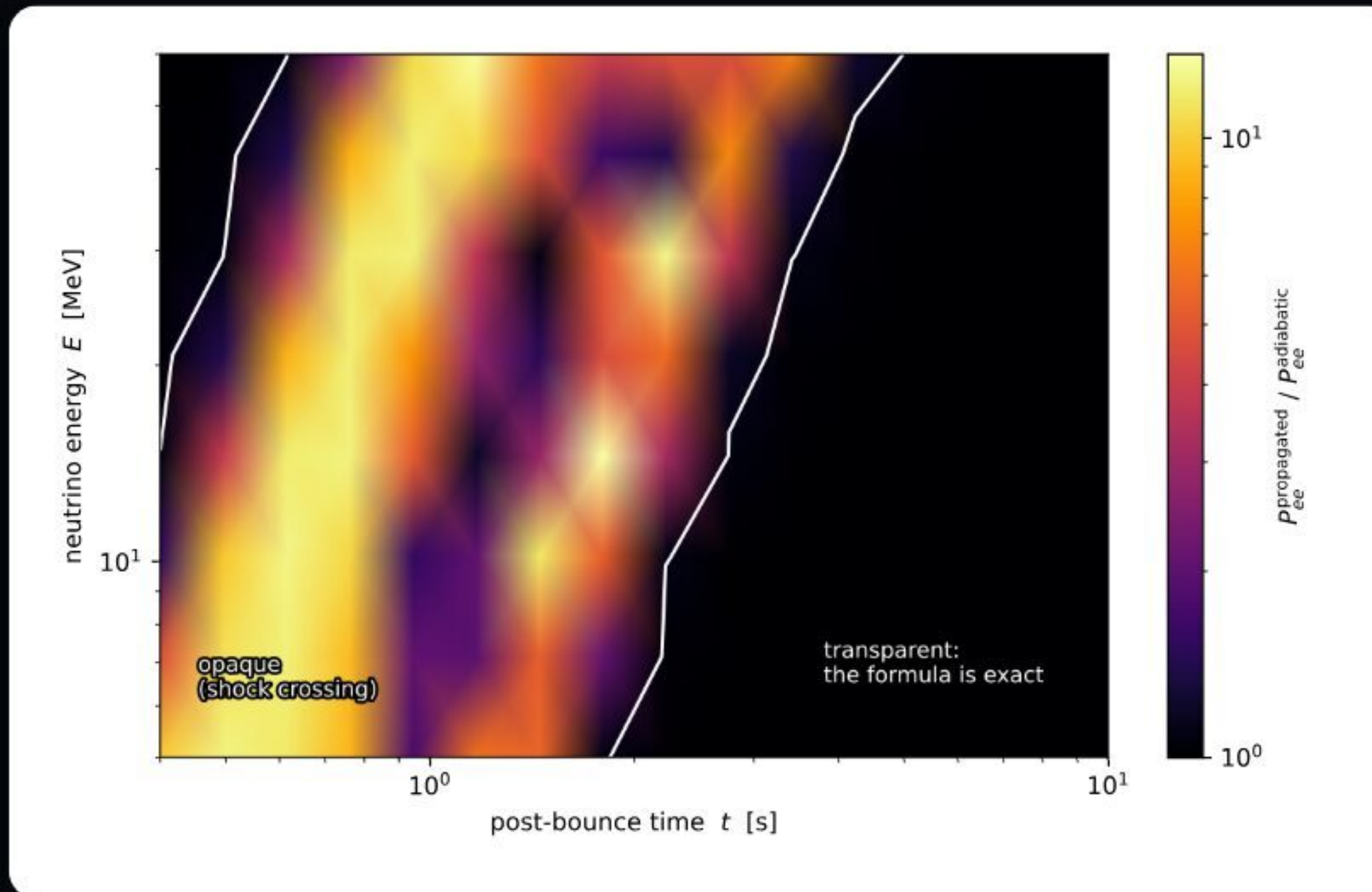
MODEL FAMILY	BIAS IN THE N_e CC COUNT
Nakazato	−4.0 to −4.2%
Garching	−3.9 and −4.4%
Sukhbold	−2.9%
Fornax 2022	+1.3 and +1.4%

SNOwGLOBES, 40 kt argon at 10 kpc.

A bias of fixed sign could be absorbed into a correction factor. One whose sign follows the flavour hierarchy of the source cannot — and it is **degenerate** with the spectral quantities one wants to extract.

It survives in both orderings, for different reasons: −4.1% in normal, −5.4% in inverted. The systematic cannot be avoided by assuming an ordering.

The error lives in a band, *in closed form*



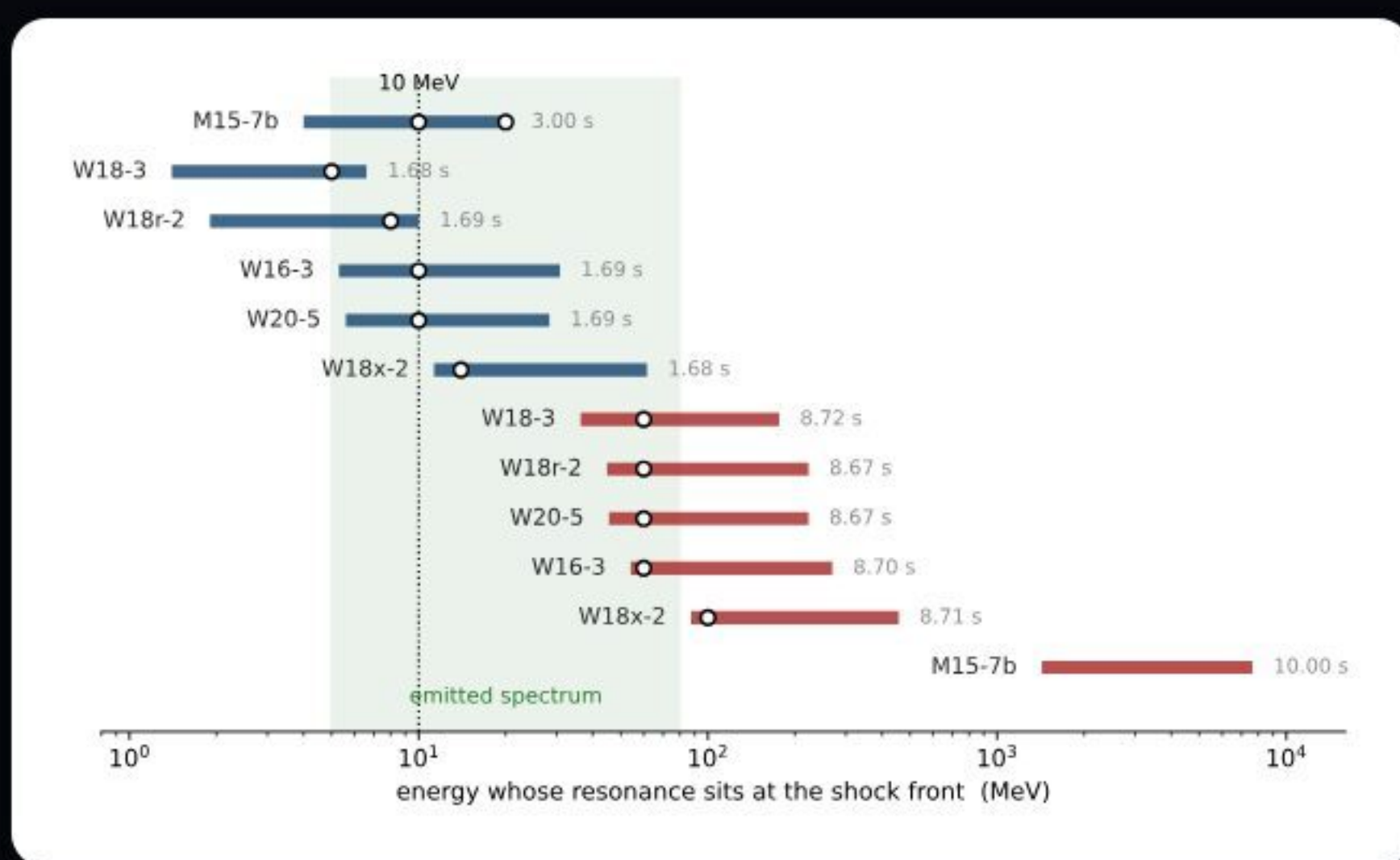
Treating the front as a constant is off by up to a factor **13.7** in a band of (t, E) that closes at $t \approx 4$ s.

The error is not spread out: it occupies a **band** in the time–energy plane, and the band has a closed form.

It follows from the presupernova density profile, given the shock radius and the compression. **With no flavour calculation at all.**

And it **closes**: past $t \approx 4$ s the shock has already left the density region the H resonance occupies at observable energies.

Every progenitor has *its own band*



Five sibling progenitors at one epoch: from **1.4 to 61.6 MeV**, with no single energy inside all of them.

Five sibling progenitors, at the same epoch, span **1.4 to 61.6 MeV**. And **no single energy** falls inside all five bands.

THE PRACTICAL CONSEQUENCE

Comparing models at a **fixed energy** returns a false null: in each model you will be outside some other model's band.

We fell into that trap before measuring it.

The band depends on each star's presupernova structure, which differs between progenitors even at the same initial mass.

What this settles *and what it doesn't*

Three statements

- **A criterion with a number:** the response is extinguished above $\approx 3 L_{\text{osc}}$. “Steepen the front by hand” becomes a checkable condition.
- **The cost:** 0.44–0.76 in P_{H} and a few per cent in the argon count, with a sign set by the source.
- **Where to look:** the (t, E) band in closed form, from presupernova structure.

Limits

- The scales that act are in no model: grids resolve nothing shorter than $4.7 L_{\text{osc}}$. The turbulence amplitude is *extrapolated*.
- Blue supergiants only (six Garching models).
- Not yet an observable: a few σ at 10 kpc, scaling as $1/d$.

R. C. R. de Lima, *Which density structures affect supernova neutrino flavour conversion, and the shock front at the MSW resonance* — in preparation (2026).

Four things to *take home*

1

Flavour oscillation is **phase interference**.

The phase is $\int p_\mu dx^\mu$ — and that holds in any geometry.

2

Supernovae and collapsars emit neutrinos **a few r_s** out.

There, redshift, bending and local quantities are not corrections: they are the physics.

3

Gravity doesn't only get in the way: with several paths, it **measures m_k^2** . DUNE and the SNEWS network are ready for the next galactic supernova.

4

In a supernova only the **shock front** passes the resonance criterion — and simulations store it too wide. Restoring it changes P_H by 0.44–0.76 and the DUNE count by a few per cent.

To go further

Premises

W. Pauli, letter of 4/12/1930
 E. Fermi, *Z. Phys.* 88, 161 (1934)
 H. Bethe, R. Peierls, *Nature* 133, 532 (1934)
 C. L. Cowan et al., *Science* 124, 103 (1956)
 G. Danby et al., *PRL* 9, 36 (1962)
 R. Davis et al., *PRL* 20, 1205 (1968)
 B. Pontecorvo, *Sov. Phys. JETP* 26, 984 (1968)
 Z. Maki, M. Nakagawa, S. Sakata, *PTP* 28, 870 (1962)
 L. Wolfenstein, *PRD* 17, 2369 (1978)
 S. Mikheyev, A. Smirnov, *Sov. J. Nucl. Phys.* 42, 913 (1985)
 Super-Kamiokande, *PRL* 81, 1562 (1998)
 SNO, *PRL* 89, 011301 (2002)
 DONUT, *PLB* 504, 218 (2001)
 NuFIT 6.0, *JHEP* 12, 216 (2024)
 KATRIN, *Science* 388, 180 (2025)

Part I

A. Einstein, *Sitzungsber. Preuss. Akad.* 844 (1915)
 K. Schwarzschild, *Sitzungsber. Preuss. Akad.* 189 (1916)
 F. Dyson, A. Eddington, C. Davidson, *Phil. Trans. A* 220, 291 (1920)
 R. Pound, G. Rebka, *PRL* 4, 337 (1960)
 R. Hulse, J. Taylor, *ApJ* 195, L51 (1975)
 LIGO/Virgo, *PRL* 116, 061102 (2016)
 EHT, *ApJL* 875, L1 (2019); 930, L12 (2022)
 S. Woosley, *ApJ* 405, 273 (1993)
 A. MacFadyen, S. Woosley, *ApJ* 524, 262 (1999)
 R. Popham, S. Woosley, C. Fryer, *ApJ* 518, 356 (1999)

Part II

K. Hirata et al., *PRL* 58, 1490 (1987)
 R. Bionta et al., *PRL* 58, 1494 (1987)
 L. Krauss, S. Tremaine, *PRL* 60, 176 (1988)
 M. Longo, *PRL* 60, 173 (1988)
 L. Stodolsky, *Gen. Rel. Grav.* 11, 391 (1979)
 N. Fornengo, C. Giunti, C. W. Kim, J. Song, *PRD* 56, 1895 (1997)
 C. Cardall, G. Fuller, *PRD* 55, 7960 (1997)
 H. Swami, K. Lochan, K. Dixit, *PRD* 102, 024043 (2020)
 M. Gasperini, *PRD* 38, 2635 (1988)
 A. Halprin, C. Leung, *PRL* 67, 1833 (1991)
 J. Beacom, R. Boyd, A. Mezzacappa, *PRD* 63, 073011 (2001)
 A. Cervera et al., *NPB* 579, 17 (2000)
 O. Caballero, G. McLaughlin, R. Surman, *ApJ* 745, 170 (2012)
 J. Cherry et al., *PRL* 108, 261104 (2012)
 DUNE FD TDR, arXiv:2002.03005 (2020)

Part III

R. Schirato, G. Fuller, astro-ph/0205390 (2002)
 G. Fogli et al., *PRD* 68, 033005 (2003)
 R. Tomàs et al., *JCAP* 0409, 015 (2004)
 V. Barger, P. Huber, D. Marfatia, *PLB* 617, 167 (2005)
 A. Dighe, A. Smirnov, *PRD* 62, 033007 (2000)
 J. Kneller, G. McLaughlin, K. Patton, *J. Phys. G* 40, 055002 (2013)
 K. Patton, J. Kneller, G. McLaughlin, *PRD* 89, 073022 (2014)
 A. Wongwathanarat, E. Müller, H.-T. Janka, *A&A* 577, A48 (2015)
 A. Baxter et al. (SNEWPY), *ApJ* 925, 107 (2022)
 M. Bustamante (NuOscProbExact), arXiv:1904.12391
 DUNE, *EPJC* 81, 423 (2021)
 R. C. R. de Lima, in preparation (2026)

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Rafael C. R. de Lima

rafael.lima@udesc.br · rafael-lima.pages.dev

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